

Highlights

Server-Anchored Trust Bootstrapping for Byzantine-Resilient and Auditable Federated Graph Learning

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- Client-declared fairness metadata captures six published FL aggregation rules.
- Falsified reports raise disparity by 0.46 DPD and cut AUC by 0.14 over 30 seeds.
- Multi-round client-wise DP drives constraint testing toward chance.
- TrustFedGNN achieves exact syntactic metadata immunity via server holdout.
- Holdout-guided alignment suppresses stealthy Byzantine fairness poisoning.

Server-Anchored Trust Bootstrapping for Byzantine-Resilient and Auditable Federated Graph Learning

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Abstract

Federated learning on graph neural networks (Federated GNNs) across distributed, privacy-sensitive environments faces a critical structural vulnerability: existing fairness-aware and contribution-based aggregation protocols invariably rely on client-reported scalar telemetry or pooled validation graphs. In this work, we demonstrate empirically that across six representative fairness-aware federated rules, malicious clients submitting falsified disparity claims capture disproportionate aggregation weights (up to 97.4% model control), severely corrupting downstream demographic parity and predictive utility. Furthermore, we establish via an information-theoretic Minimax lower bound (Le Cam’s two-point method with Pinsker’s inequality) that privatising self-reported statistics via Local Differential Privacy (LDP) inevitably collapses server observability ($\mathcal{R}^* \geq 43.5\%$ at $\epsilon = 8.0$, rising to $\geq 49.0\%$ at $\epsilon = 1.0$) due to perpetual folded-normal bias. To resolve this structural impasse, we propose TRUSTFEDGNN, a server-anchored federated graph learning framework operating under strict Kerckhoffs security assumptions. By deriving a dual-objective reference vector on an isolated server holdout graph (D_{root}) and filtering client updates via non-linear coordinate gating (EMA smoothing, non-linear rectification, and dynamic norm rescaling), TRUSTFEDGNN establishes exact syntactic metadata immunity by construction ($\Delta w \equiv 0$). Across extensive evaluations on four real-world

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graph benchmarks (German Credit, Bail Recidivism, Credit Default, and Pokec-z) under three tiers of adversarial attacks, TRUSTFEDGNN restricts stealth adversaries below equal share ($0.68\times$ – $0.73\times$) without imposing any robustness tax on group fairness. Finally, TRUSTFEDGNN incorporates an auditable, linear-complexity first-order contribution decomposition $O(KP)$ for per-round regulatory compliance logging, paired with a multi-criteria governance ranking whose stability is empirically preserved across 2,000 Monte Carlo weight configurations ($\rho_s = 0.965$), effectively bridging robust federated graph learning with enterprise risk governance.

Keywords: Federated Learning, Graph Neural Networks, Byzantine Resilience, Group Fairness, Minimax Lower Bound, Auditable AI

1. Introduction

Distributed machine learning on relational networks, structured as Federated Graph Neural Networks (FedGNNs), has emerged as a cornerstone paradigm for privacy-preserving collaborative intelligence across regulated domains, including anti-money laundering in financial consortia, legal recidivism risk assessment, and clinical healthcare networks [1, 2, 3, 4]. In these cross-silo enterprise federations, graph structures encode sensitive relationships between legal entities, bank accounts, or patient cohorts. Legal mandates, such as the European Union AI Act, the General Data Protection Regulation (GDPR), and the U.S. Equal Credit Opportunity Act, strictly forbid centralizing raw relational data across sovereign institutional boundaries. Concurrently, these statutes demand that predictive algorithms ensure non-discriminatory group demographic fairness while maintaining reliable operation in the presence of malicious or compromised participant nodes.

The Trustworthy Federated Learning Trilemma. A fundamental architectural challenge in decentralized graph learning is that the system must reconcile three inherently competing objectives:

1. **Unconstrained Task Utility (U):** Maximizing predictive accuracy by solving $\min_{\theta} \mathcal{L}_{\text{task}}(\theta)$. Standard single-objective aggregation protocols (e.g., FedAvg, FLTrust) dedicate the entirety of model parameter updates to minimizing empirical task loss, reaching high accuracy on clean data. However, because they allocate zero gradient capacity to algorithmic fairness, they leave demographic disparity unconstrained ($\text{DPD} \geq 0.055$ on

German Credit) and remain blind to adversarial perturbations injected into the fairness subspace.

2. **Group Demographic Fairness (F):** Enforcing demographic parity ($\text{DPD}(\theta) \leq \tau$) across protected populations. Existing fairness-aware federated protocols (e.g., FairFed, $F^2\text{GNN}$, FedGraph-Fair, BFWA) rely on client-reported scalar disparity metrics to coordinate aggregation weights.
3. **Byzantine Robustness (R):** Bounding the influence of arbitrary or malicious clients ($w_{\text{adv}} \leq 1/K$). Standard robust geometric filters (e.g., Coordinate Median, Krum, FLAME) assume symmetric parameter distributions, treating outlier vectors as anomalies.

In decentralized relational learning, these three objectives form the *Trustworthy FL Trilemma*: unconstrained optimization along any one or two axes inevitably induces structural failure on the remaining dimensions.

Failure Mode 1: The Self-Reporting Metadata Trap. The dominant design pattern in contemporary fairness-aware federated learning coordinates client aggregation weights based on self-reported scalar telemetry, such as local demographic disparity $\widehat{\text{DPD}}_k$ or local empirical loss [5, 6, 7, 8], including our own earlier conference formulation [9]. In adversarial environments governed by Kerckhoffs’ principle, this reliance creates a critical attack surface. As established in our empirical vulnerability evaluation (Section 4.2, Table 2), Byzantine adversaries can transmit poisoned gradient updates while falsifying local disparity claims ($\widehat{\text{DPD}}_k \approx 0.000$). Across six distinct self-reporting aggregation rules, adversaries systematically hijack the weighting mechanism: in $F^2\text{GNN}$, the attacker captures $w_{\text{adv}} = 0.9737$ of global model authority ($4.87\times$ its uniform share), inflating downstream disparity by $\Delta\text{DPD} = +0.4638$ ($p = 1.9 \times 10^{-9}$). Client-reported metadata interfaces are fundamentally fragile under adversarial incentives.

Failure Mode 2: The Information-Theoretic Barrier of Local Differential Privacy. A natural defense is to mask self-reported scalar metrics via Local Differential Privacy (LDP). However, applying noise to disparity statistics encounters two insurmountable theoretical barriers:

1. *Perpetual Expectation Bias (Lemma 1):* Because disparity is evaluated via an absolute difference operator ($\widehat{\text{DPD}}_k = |\tilde{\mu}_{k,0} - \tilde{\mu}_{k,1}| \geq 0$), zero-

mean Gaussian noise is transformed into a central folded-normal distribution [10]. Conditioned on past rounds $\mathcal{F}_{t-1}$, the expectation remains strictly positive ($\mathbb{E}[\widehat{\text{DPD}}_k] = \tilde{\sigma}_k \sqrt{2/\pi} > 0$), and almost surely $\sum_k w_k \widehat{\text{DPD}}_k \geq \min_k \widehat{\text{DPD}}_k > 0$. The noise cannot cancel across silos, dilating the apparent disparity budget by up to +10,891% (Table 3).

2. *Minimax Observability Barrier (Theorem 4)*: Applying Le Cam’s two-point method and Pinsker’s inequality, we prove that distinguishing fair from unfair states under LDP is bounded from below by minimax testing risk $\mathcal{R}^* \geq 43.5\%$ at privacy budget $\epsilon = 8.0$ (rising to $\geq 49.0\%$ at $\epsilon = 1.0$, Table 3). By the Data Processing Inequality, no server-side post-processing can lower this observational floor.

Furthermore, while applying DP-SGD across the full high-dimensional parameter tensor reduces membership leakage, it destroys predictive utility, incurring a parameter degradation penalty 115 $\times$ more severe than scalar-channel privacy (Figure 2).

Failure Mode 3: Geometric Defense Failures and Enterprise Constraints. Turning to classical Byzantine defenses reveals equally severe structural vulnerabilities:

1. *Fairness Subspace Blindness (Theorem 5)*: Anchor-free geometric detectors (e.g., Krum, FLAME) operate over the cosine geometry of task parameter updates. Because task classification coordinates dominate the gradient spectrum by orders of magnitude relative to low-rank fairness parameters ($\mathcal{D}_{\text{cos}} \approx \mathcal{D}_{\text{cos}}^{\text{task}}$), proximity-stealth adversaries can inject substantial fairness degradation while remaining geometrically indistinguishable inside the benign clustering ball ($\rho = 0.85$).
2. *The Shielding Paradox (Median Backfire)*: On non-IID graph federations with modular community partitions, benign clients in minority topological clusters naturally exhibit elevated gradient variance. Symmetric coordinate-wise median filtering systematically purges benign minority gradients before eliminating stealth adversaries. As demonstrated in Section 4.3 (Table 5), median filtering backfires, inflating adversary aggregation share by +82.9% relative to canonical averaging ($w_{\text{adv}} = 0.5343$ at $f/K = 0.4$).

Finally, heavyweight cryptographic mechanisms (e.g., ZK-SNARKs, Secure MPC) and hardware enclaves (TEEs) are practically prohibitive in cross-silo graph federations due to extreme graph message-passing computational overhead and heterogeneous enterprise infrastructure.

The TRUSTFEDGNN Paradigm: Exogenous Server Anchoring. Eliminating these flawed avenues forces a necessary conclusion: robust, fair aggregation cannot depend on client-reported metadata or symmetric unanchored geometry. Instead, the server must evaluate incoming parameter deltas against an independent, physically isolated directional reference.

We propose TRUSTFEDGNN, an auditable cross-silo framework that achieves balanced Pareto operation across the Trustworthy FL Trilemma. TRUSTFEDGNN implements a principled two-tier decoupling:

- **Client Side (FTGD):** Local silos execute Fair Targeted Gradient Descent with first-order orthogonal projection (Theorem 1), filtering sensitive demographic gradients locally prior to parameter transmission.
- **Server Side (Canonical FU-Alignment):** The server maintains an isolated audit anchor dataset D_{root} (consisting of 125 stratified nodes) to compute an exogenous dual-objective target gradient $g_{\text{target}} = g_{\text{task}}^{\text{srv}} + \alpha g_{\text{fair}}^{\text{srv}}$. Canonical aggregation combines rectified directional cosine gating, dynamic norm rescaling, and exponential moving average (EMA) smoothing. This mathematical formulation achieves exact syntactic metadata immunity ($\Delta w = 0.0000$, Theorem 2) while dampening round-to-round weight turbulence. To mitigate crude unbounded scaling attacks ($c = 100$) without inducing the shielding paradox under stealth, a defense-in-depth coordinate median screening variant (`robust_fu_shapley`) is provided as a static configuration arm.

Key Scientific Contributions. This work delivers four primary contributions to trustworthy decentralized graph learning:

1. **Formalization of the Trustworthy FL Trilemma and Threat Hierarchy:** We formalize the structural conflict between Task Utility (U), Demographic Fairness (F), and Byzantine Robustness (R) in decentralized graph learning. Under Kerckhoffs’ principle, we formulate a 3-tier adversarial hierarchy and prove exact syntactic metadata immunity ($\Delta w = 0.0000$) as an architectural invariant (Theorem 2), categorizing aggregation rules into naive scalar consumers, endogenous references, and exogenous anchors.

2. **Information-Theoretic Limits and Subspace Blindness:** We prove the perpetual expectation bias of folded-normal perturbations under LDP (Lemma 1) and establish the Le Cam minimax observability lower bound $\mathcal{R}^* \geq 43.5\%$ at $\epsilon = 8.0$ (rising to $\geq 49.0\%$ at $\epsilon = 1.0$, Theorem 4, Table 3). Furthermore, we prove Fairness Subspace Blindness for anchor-free geometric detectors (Theorem 5), explaining why clustering algorithms fail against proximity-stealth fairness attacks.
3. **Two-Tier Architecture and Formal Axiomatic Verification:** We propose TRUSTFEDGNN, integrating local FTGD orthogonal projection with server-side Canonical FU-Alignment. To ensure strict formal correctness, Theorem 1 (first-order orthogonality) and Propositions 1–3 (first-order linear decomposition in $O(KP)$, simplex validity, and null-player guarantee) are formally verified with machine-checked proof certificates.
4. **Empirical Verification and Auditable Governance:** Across four real-world graph benchmarks (Pokey-z, German Credit, Bail Recidivism, Credit Default), 14 baselines, and three attack tiers, we demonstrate that TRUSTFEDGNN establishes an intentional, balanced operating point on the Pareto frontier (trading 2.00% AUC margin for a 41.5% disparity reduction on Pokey-z, while achieving statistical equivalence on benign data: $\Delta\text{AUC} = +0.0015, p = 0.625$). We provide an $O(KP)$ per-round governance ledger ($r = 0.7662 < 0.80$ against exponential combinatorial Shapley), and transparently report operational boundaries under extreme Dirichlet non-IID shifts and white-box memory compromises.

Paper Organization. The remainder of this manuscript is organized as follows. Section 2 positions our framework within five foundational literature paradigms and outlines operational boundaries. Section 3 presents the formal problem formulation, the TRUSTFEDGNN architecture, theoretical observability bounds, and formal derivations of governance properties. Section 4 reports extensive empirical stress tests, SOTA benchmark comparisons, ablation studies, and causal attribution analyses. Section 5 concludes the paper with operational deployment guidelines and research limitations.

2. Related Work and Intellectual Lineage

Decentralized machine learning on relational structures occupies the intersection of graph representation learning, algorithmic fairness, and distributed security. In this section, we organize prior literature into five distinct

paradigms, characterize their foundational failure modes under adversarial stress, and delineate the operational scope of TRUSTFEDGNN.

2.1. Paradigms I & II: Decentralized Graph Learning and In-Processing Debiasing

2.1.1. Graph Neural Networks in Federated Settings (Paradigm I)

Federated Graph Neural Networks (FedGNNs) extend decentralized optimization [1, 4] to non-Euclidean domains, enabling multi-institution collaboration across cross-silo topologies without centralizing sensitive edge structures [2, 3]. Early baseline protocols primarily employ unconstrained coordinate averaging, such as FedAvg-GCN, to minimize empirical task risk across isolated client subgraphs. While establishing a valuable reference ceiling for unconstrained predictive accuracy, Paradigm I protocols neglect demographic disparity entirely. In regulated domains such as credit assessment and legal risk scoring, neighborhood aggregation propagates and amplifies historical biases embedded in graph connectivity, yielding severe group disparity.

2.1.2. Client-Side In-Processing Debiasing (Paradigm II)

To mitigate demographic disparity on graphs, a substantial body of research develops client-side in-processing mechanisms. Representative frameworks include adversarial feature disentanglement in FairGNN [11], sensitive neutralization in FairSIN [12], topology re-balancing in FairGB [13], invariant learning in FairInv [14], data debiasing in EDITS [15], fair contrastive learning in NIFTY [16], disentangled representation learning in DAB-GNN [17], distribution-aware fairness in FnRGNN [18], and vertical federated fairness in FAVGNN [19].

Aggregation Boundary Blind Spot. While Paradigm II methods achieve localized bias reduction on benign graphs, their operational boundary is strictly confined to client-internal training. At the server aggregation boundary, these algorithms remain entirely unprotected. Standard parameter averaging at the server treats debiased updates from honest silos identically to poisoned updates submitted by Byzantine adversaries. When adversarial silos inject malicious gradients or manipulate node labels, localized in-processing mechanisms provide zero recourse against global model subversion.

2.2. *Paradigm III: Client-Reported Metric Coordination and The Metadata Trap*

2.2.1. *Heuristic and Lagrangian Disparity Coordination*

Recognizing the limits of static averaging, Paradigm III adapts server aggregation weights using telemetry transmitted by client silos. Frameworks such as FairFed [5] and FairGFL [6] employ heuristic reweighting based on deviations between local and global fairness metrics. Lagrangian coordination protocols such as BFWA [9]—our earlier preliminary conference baseline—and constrained optimization schemes such as q -FedAvg [20] and PUFFLE [21] formulate federated fairness as dual subgradient or Frank–Wolfe optimization, dynamically adjusting client coefficients to steer global demographic disparity below a predetermined budget τ . Recent extensions include controllable calibration in FedFACT [22] and differentially private fairness coordination in FedFDP [23]. Similarly, structure-aware frameworks such as F^2 GNN [7] and FedGraph-Fair [8] incorporate relational metrics and topological fairness statistics into client weighting.

2.2.2. *The Structural Flaw of Self-Reporting: Interface Manipulation*

The fatal vulnerability shared by all Paradigm III methods lies in their reliance on unverified, client-reported scalar metadata ($\widehat{\text{DPD}}_k$, local loss values, or subgroup demographic counts). In decentralized federations governed by Kerckhoffs’ principle, malicious clients can falsify scalar telemetry while transmitting poisoned model updates. As proven by our syntactic metadata analysis (Theorem 2) and demonstrated across empirical vulnerability audits (Section 4.2), adversaries reporting fabricated parity metrics can unilaterally capture a dominant fraction of the global aggregation weight, systematically undermining downstream fairness while destabilizing optimization. Furthermore, nominal mitigation strategies such as polynomial dampening in PoPETs-FairFed [24] merely attenuate update magnitudes rather than providing structural immunity against semantic metadata manipulation.

2.3. *Differential Privacy in Graph FL and Information-Theoretic Barriers*

2.3.1. *Local Differential Privacy and the Privacy–Fairness Tension*

A natural defense against metadata manipulation and membership leakage is Local Differential Privacy (LDP) [25, 26]. However, extensive literature documents a fundamental tension between differential privacy and demographic fairness [27, 28, 29, 30], wherein private perturbation disproportionately harms minority demographic groups.

2.3.2. Irreducible Folded-Normal Expectation Bias (Lemma 1)

When LDP perturbation is applied to client-reported disparity statistics $\widetilde{\text{DPD}}_k = |\tilde{\mu}_{k,0} - \tilde{\mu}_{k,1}|$, the absolute value operator induces an irreducible expectation bias. As established in Lemma 1, because the transformation maps zero-mean Gaussian noise onto a central folded-normal distribution [10], the expectation conditioned on past rounds remains strictly positive:

$$\mathbb{E} \left[\sum_{k=1}^K w_k \widetilde{\text{DPD}}_k \mid \mathcal{F}_{t-1} \right] = \sqrt{\frac{2}{\pi}} \sum_{k=1}^K w_k \tilde{\sigma}_k > 0. \quad (1)$$

Because disparity is strictly non-negative ($\widetilde{\text{DPD}}_k \geq 0$), the noise cannot cancel through cross-client averaging. In multi-round federated training, this cumulative positive bias artificially dilates the apparent constraint violation by multiple orders of magnitude, precluding accurate dual coordination.

2.3.3. The Minimax Le Cam Observability Barrier (Theorem 4)

Beyond expectation bias, Theorem 4 establishes an information-theoretic lower bound on the hypothesis testing risk of distinguishing fair from unfair states under LDP. Applying Le Cam’s two-point method [31] and Pinsker’s inequality [32], the server minimax classification error is bounded from below:

$$\mathcal{R}^* \geq \frac{1}{2} \left(1 - \frac{\tau \sqrt{K}}{2\tilde{\sigma}} \right). \quad (2)$$

By the Data Processing Inequality, no algorithmic post-processing or adaptive reweighting on the server can overcome this fundamental observational barrier.

2.3.4. Two-Channel Leakage Separation

Crucially, our formulation resolves this privacy bottleneck through architectural channel separation. Applying DP-SGD across the entire parameter tensor incurs a catastrophic utility penalty. In contrast, restricting differential privacy exclusively to the low-dimensional scalar reporting channel allows rigorous local auditing without contaminating model gradient representations (Section 4.2).

2.4. Paradigm IV: Geometric Byzantine Filtering and The Shielding Paradox

2.4.1. Coordinate-Wise and Distance-Based Aggregation

To defend against Byzantine workers, classical robust distributed learning relies on geometric outlier filtering without external reference data. Prominent paradigms include coordinate-wise median and trimmed mean [33], distance-based neighbor screening such as Krum and Multi-Krum [34], multi-round robust pruning in Bulyan [35], angular similarity clustering in Fools-Gold [36], information-theoretic aggregation in BYIT-FL [37], and density-based spectral clustering in FLAME [38].

2.4.2. Fairness Subspace Blindness (Theorem 5)

While effective against crude arithmetic poisoning (e.g., sign-flipping and isotropic Gaussian noise), anchor-free geometric detectors suffer from a structural blind spot. Targeted model poisoning aimed at algorithmic fairness has recently emerged as an acute threat, demonstrated by bias amplification in EAB-FL [39] and fairness-constrained optimization attacks [40]. As formalized in Theorem 5, because pairwise cosine and Euclidean statistics operate over the full parameter space, adversaries can construct perturbations orthogonal to the task gradient ($\delta_{\text{fair}} \in \mathcal{S}_{\text{fair}} \perp \mathcal{S}_{\text{task}}$) that remain strictly within the benign clustering radius ($\mathcal{D}_{\text{cos}} < \rho_{\text{thresh}}$). In deep GNN architectures, task-specific parameters overwhelmingly dominate the parameter spectrum relative to the low-rank fairness subspace ($\mathcal{D}_{\text{cos}} \approx \mathcal{D}_{\text{cos}}^{\text{task}}$). Consequently, detectors such as FLAME remain blind to targeted fairness degradation injected by proximity-stealth adversaries (Section 4.3).

2.4.3. The Shielding Paradox on Heterogeneous Graph Topologies

On non-IID graph federations characterized by modular community structure, client parameter updates naturally exhibit high directional variance reflecting local topological homophily. Symmetrical geometric filters systematically conflate legitimate topological diversity with malicious deviation. Under proximity stealth poisoning, coordinate-wise median filtering purges benign minority gradients before eliminating camouflage adversaries. This induces the *Shielding Paradox (Median Backfire)*, where robust filtering inadvertently inflates the relative influence of the adversary compared to uniform averaging (Section 4.3).

2.5. Paradigm V: Server-Anchored Alignment and Contribution Accounting

2.5.1. Endogenous vs. Exogenous Reference Anchoring

To overcome the limitations of unanchored filtering, Paradigm V introduces reference-based trust bootstrapping:

- **Endogenous Reference Anchoring (Class b):** Frameworks such as CGSV [41] evaluate client contributions against a reference update computed from the participating client consortium (e.g., via pooled validation subsets or consensus vectors). While achieving syntactic metadata immunity, endogenous references remain vulnerable to poison accumulation: coordinated Byzantine coalitions can skew the reference direction itself.
- **Exogenous Reference Anchoring (Class c):** Frameworks such as FLTrust [42], GuardFed [43], and FLTG [44] bootstrap trust by evaluating client updates against an independent, server-maintained holdout dataset D_{root} . Because the reference gradient is evaluated outside the client consortium, the directional compass is physically isolated from adversarial manipulation.

2.5.2. Single-Objective Task Tunnel Vision

Most exogenous protocols align client updates exclusively with the server task gradient $g_{\text{task}}^{\text{srv}}$ ($\alpha = 0$), dedicating zero capacity to group fairness and leaving the model unprotected against adversarial fairness drift; FLTrust is the canonical instance. Furthermore, raw cosine weighting induces severe round-to-round weight volatility, destabilizing convergence on non-IID topologies. A recent exception is GuardFed [43], which augments the server reference with a fairness-aware component and screens clients by a dual-perspective trust score. Our contribution is therefore not the observation that a server anchor may carry a fairness term, but three properties that a threshold-screening dual reference does not provide: (i) exact syntactic metadata immunity as a domain invariant rather than an empirical robustness result (Theorem 2); (ii) the endogenous/exogenous anchor taxonomy that explains why closing the declared channel is necessary but not sufficient (Remark 3); and (iii) an $O(KP)$ per-round attribution ledger with machine-checked axiomatic properties (Propositions 1–3). We further separate the fairness term from the robustness mechanism and report it as mechanically inert under our attack model (Section 4.4) rather than as a source of robustness.

2.5.3. Cooperative Game Theory and the Shapley Bottleneck

Evaluating individual client contributions via cooperative game theory has received substantial attention, notably through GTG-Shapley [45] and recent axiomatic frameworks [46, 47]. However, exact Shapley computation requires evaluating model utility across all 2^K client coalitions, while Monte Carlo approximations require hundreds of retraining steps per round. In enterprise graph federations with communication time constraints, exponential game-theoretic evaluation is computationally intractable. In Section 3.4, Proposition 1 resolves this bottleneck by establishing an $O(KP)$ first-order linear decomposition, providing an efficient governance logging mechanism while maintaining transparent delineation from combinatorial Shapley values.

2.6. Operational Demarcation and Enterprise Practicality Scope

2.6.1. Exclusion of Heavy Cryptographic Overhead in Cross-Silo Graph FL

In cross-silo federations, several cryptographic and economic approaches are theoretically appealing but practically prohibitive:

1. **Zero-Knowledge Proofs (ZK-SNARKs):** Generating cryptographic proofs for sparse graph message passing across multi-hop neighborhoods incurs prohibitive computational overhead (minutes per step and large proof payloads), violating enterprise service-level agreements.
2. **Secure Multi-Party Computation (SMPC):** Secret-sharing multi-hop graph embeddings across wide-area networks creates severe communication bottlenecks scaling quadratically with participant count.
3. **Hardware Enclaves (TEEs):** Legacy server diversity across enterprises precludes uniform enclave assumptions; furthermore, GPU enclaves remain nascent in production.
4. **Peer Prediction and Repeated Games:** Economic incentive schemes assume rational participants; they degrade rapidly under arbitrary or non-rational Byzantine adversaries seeking maximal model destruction.

2.6.2. Architectural Decoupling in TRUSTFEDGNN

To navigate these constraints, TRUSTFEDGNN implements a clear operational decoupling:

- **Client Side:** Local silos execute Fair Targeted Gradient Descent (FTGD) with first-order orthogonal projection (Theorem 1) to decouple task optimization from local sensitive attributes.
- **Server Side:** The server maintains an isolated, stratified audit anchor D_{root} (125 nodes), executing Canonical FU-Alignment (`fu_shapley`) with rectified cosine gating, dynamic norm rescaling, and exponential smoothing. To protect against unbounded scaling without inducing the shielding paradox under camouflage, defense-in-depth coordinate median screening (`robust_fu_shapley`) is provided as a static configuration variant for high-threat deployments.

2.6.3. Demarcation from Preliminary Conference Work

An earlier preliminary conference version of this investigation appeared as FedFairGNN [9] (PMLR vol. 319, pp. 74–86, IndabaX Nigeria 2026), which focused on client-side regularized message passing (FSER), local gradient projection (FTGD), and the heuristic BFWA aggregation rule for fraud detection across three clients ($K = 3$). To maintain rigorous scientific demarcation and transparent provenance, we explicitly delineate the components inherited versus the novel contributions established in the present journal expansion:

1. **Inherited Client-Side Modules:** Local Fair Targeted Gradient Descent (FTGD) is inherited from [9] as a client-side in-processing mechanism, where its theoretical scope is narrowed to exact linear subspace orthogonality (Theorem 1). Client-side attention regularization (FSER) is retained strictly as an internal baseline, with ablation revealing a null effect on pure topology graphs (Table 11).
2. **Self-Auditing and Re-Positioning of BFWA:** In [9], BFWA adjusted server aggregation coefficients based on client-reported performance and disparity scalars. Rather than treating BFWA as an external straw man, the present work subjects our own preliminary rule to rigorous adversarial audit on identical footing (Section 4.2, Table 2). We prove that BFWA is captured at $w_{\text{adv}} = 0.5703$ ($2.85\times$ uniform share), demonstrating that the vulnerability is structural to the self-reporting interface rather than an optimization artifact (Theorem 3).
3. **Architectural Paradigm Shift (FU-Alignment):** Completely absent from [9], the canonical server-side FU-Alignment aggregator (`fu_shapley`)

replaces unverified client declarations with an exogenous holdout audit anchor D_{root} , achieving exact syntactic metadata immunity ($\Delta w \equiv 0$, Theorem 2) alongside scaling-safe norm rescaling.

4. **Information-Theoretic and Formal Proof Suite:** The minimax observability lower bound under LDP (Theorem 4), the folded-normal expectation bias (Lemma 1), and the formal axiomatic derivation of contribution properties (Propositions 1–3) are entirely novel theoretical contributions of this manuscript.
5. **Expanded Empirical Validation:** The experimental scope expands from three clients on fraud subgraphs to comprehensive stress testing across six benchmark datasets (social networks, recidivism, credit default), multi-ratio Byzantine poisoning, Dirichlet topology skew, and 30 paired random seeds per contrast.

3. Methodology

This section presents the formal methodology of TRUSTFEDGNN. We begin by formulating the decentralized graph federated learning setting, introducing the *Trustworthy FL Trilemma*, and defining our threat model under Kerckhoffs’ principle (§3.1). We then detail the TRUSTFEDGNN architecture, encompassing client-side Fair Targeted Gradient Descent (FTGD), directional server anchoring, and our decoupled two-tier aggregation rule (§3.2). Next, we establish the theoretical foundations of the framework, including exact syntactic metadata immunity, the folded-normal bias lemma, baseline dual convergence, the information-theoretic minimax observability lower bound under Local Differential Privacy (LDP), and fairness subspace blindness (§3.3). Finally, we formalize an auditable first-order contribution decomposition for enterprise governance with machine-checked axiomatic properties (§3.4).

3.1. Problem Formulation and the Trustworthy FL Trilemma

3.1.1. Decentralized Graph Federated Learning Formulation

Let $\mathcal{G} = (\mathcal{V}, \mathcal{E}, X, A)$ denote a global attributed graph partitioned across K decentralized institutional client silos $\mathcal{S}_1, \dots, \mathcal{S}_K$. Each client silo $k \in [K]$ possesses a private, non-overlapping local subgraph $\mathcal{G}_k = (\mathcal{V}_k, \mathcal{E}_k, X_k, A_k)$, where $\mathcal{V}_k \subset \mathcal{V}$ represents local entity nodes, $\mathcal{E}_k \subset \mathcal{E}$ denotes intra-silo edges,

$X_k \in \mathbb{R}^{|\mathcal{V}_k| \times d}$ is the node feature matrix, and $A_k \in \{0, 1\}^{|\mathcal{V}_k| \times |\mathcal{V}_k|}$ represents the local adjacency matrix. Cross-silo inter-graph edges $\mathcal{E}_{\text{cross}} = \mathcal{E} \setminus \bigcup_{k=1}^K \mathcal{E}_k$ remain unobserved by individual clients due to commercial secrecy and regulatory isolation.

Each node $v \in \mathcal{V}_k$ is associated with a ground-truth binary task classification label $Y_k(v) \in \{0, 1\}$ (e.g., credit default risk or loan delinquency) and a protected demographic indicator $S_k(v) \in \{0, 1\}$ (e.g., gender, age category, or racial group). The federated consortium seeks to collaboratively optimize the shared parameters $\theta \in \mathbb{R}^P$ of a Graph Neural Network (GNN) backbone to minimize the global classification risk while enforcing strict algorithmic fairness across demographic subgroups. Specifically, we focus on Demographic Parity Disparity (DPD), defined as the absolute difference in positive prediction rates between protected groups:

$$\text{DPD}(\theta) = \left| P(\hat{Y} = 1 \mid S = 0) - P(\hat{Y} = 1 \mid S = 1) \right|, \quad (3)$$

subject to the operational demographic parity constraint:

$$\text{DPD}(\theta) \leq \tau, \quad (4)$$

where $\hat{Y} = f_\theta(\mathcal{G})$ is the binary classification outcome and $\tau \geq 0$ denotes the fairness tolerance. In secondary analyses, we also monitor Equal Opportunity Disparity (EOD), which constrains disparity conditioned on positive ground-truth labels:

$$\text{EOD}(\theta) = \left| P(\hat{Y} = 1 \mid Y = 1, S = 0) - P(\hat{Y} = 1 \mid Y = 1, S = 1) \right|. \quad (5)$$

3.1.2. The Trustworthy FL Trilemma

A fundamental challenge in decentralized graph learning is that the system must satisfy three inherently competing desiderata:

1. **Unconstrained Task Utility (U):** Maximizing predictive accuracy by solving $\min_\theta \mathcal{L}_{\text{task}}(\theta)$. Single-objective baselines dedicate the entire parameter space to task loss minimization, reaching high clean-data utility (e.g., FLTrust). However, because they allocate zero gradient capacity to demographic parity, they leave group disparity unconstrained and fail to defend against adversarial perturbations directed into the fairness subspace (Section 4.3).

2. **Group Demographic Fairness (F):** Enforcing demographic parity ($\text{DPD}(\theta) \leq \tau$). Existing fairness-aware federated rules (such as FairFed, $F^2\text{GNN}$, FedGraph-Fair, and BFWA) rely on client-reported scalar meta-data ($\widehat{\text{DPD}}_k$). In adversarial settings, this creates a critical vulnerability: malicious clients reporting falsified claims can capture a dominant share of global aggregation weights, driving downstream disparity to severe levels (Section 4.2).
3. **Byzantine Robustness (R):** Bounding adversarial influence on parameter updates ($w_{\text{adv}} \leq 1/K$). Standard Byzantine-resilient filters (e.g., Coordinate Median, Trimmed Mean, Krum) rely on symmetric distributional assumptions. On heterogeneous, non-IID graph topologies, these coordinate-wise operators conflate legitimate minority graph clusters with geometric outliers. This induces the *Shielding Paradox (Median Backfire)*: symmetric median screening purges benign minority gradients before suppressing proximity-stealth adversaries, paradoxically inflating adversarial weight share (Section 4.3).

Remark 1 (Pareto Operating Point Positioning). *Simultaneously maximizing all three objectives is impossible in the general setting. Rather than asserting unconstrained dominance across all three dimensions, TRUSTFEDGNN establishes an intentional, balanced operating point on the multi-objective Pareto frontier: trading a small task utility margin for substantial reduction in group demographic disparity, while maintaining exact syntactic metadata immunity and dynamic weight adaptability without imposing a robustness tax on fairness (quantitative trade-offs are detailed in Section 4.3).*

3.1.3. Kerckhoffs Threat Model and Attack Hierarchy

We formulate our threat model under Kerckhoffs’ security principle: the adversary possesses full knowledge of the system architecture, loss formulations, and server aggregation algorithms. Up to f out of K silos ($f < K/2$) are Byzantine adversaries executing coordinated attacks across three hierarchical tiers:

1. **Tier 1 (Crude Arithmetic Poisoning):** Malicious silos execute blind parameter manipulation without access to benign gradient distributions or server data, including sign-flipping ($\Delta_k \leftarrow -c\Delta_k$), Gaussian noise perturbation, and gradient norm amplification ($c = 100$). This tier also

encompasses fairness poisoning sweeps under task-neutral budgets $\varepsilon \in [0.005, 0.10]$.

2. **Tier 2 (Adaptive Proximity Stealth):** Adversaries observe benign parameter distributions and craft poisoned updates confined within the benign coordinate-wise median radius ($\rho = 0.85$), while concurrently falsifying local disparity claims ($\widehat{\text{DPD}}_k = 0.000$).
3. **Tier 3 (White-Box Alignment Adversary):** An omniscient adversary possessing direct read access to the server directional reference g_{target} , executing projected gradient ascent (PGD, 50 steps) to track the reference direction while injecting orthogonal fairness degradation.

Pre-emptive Threat Boundary and Physical Isolation. In accordance with Kerckhoffs’ principle, an omniscient adversary with internal read access to the server reference gradient g_{target} (Tier 3 probe H_{T1}) represents the physical security limit of reference-based defense (as quantified in Section 4.3). In decentralized cross-silo financial and healthcare federations, zero-knowledge proofs (ZK-SNARKs) and hardware enclaves (TEEs) incur significant verification overhead, while peer-prediction schemes degrade under collusion. Consequently, physical server memory isolation of the audit anchor D_{root} constitutes the essential, verifiable security perimeter of the framework.

3.2. The TRUSTFEDGNN Framework and Two-Tier Architecture

3.2.1. Structural Decoupling: GNN Representation vs. Aggregation Rule

To prevent confounding architectural inductive bias with aggregation mechanics, we decouple the system into two distinct components:

1. **Representation Backbone:** A 2-layer GAT [3] with multi-head attention, batch normalization, and skip-connections on local subgraphs $\mathcal{G}_k$.
2. **Aggregation Rule:** Canonical FU-Alignment, operating on client parameter deltas $\Delta_k^{(t)} = \theta_k^{(t)} - \theta^{(t)}$ at the server.

As established in our structural control experiments (Section 4.3), the empirical utility gain observed over standard FedAvg-GCN represents a system-level integration gain driven predominantly by the representation backbone (a deep GAT scaffold with batch normalization, skip connections, and FTGD). Crucially, when evaluating the aggregation rule in isolation on an identical GAT backbone (RUN-CTRL), canonical FU-Alignment achieves statistical

parity with standard FedAvg on benign data (two one-sided test equivalence within margin $\delta = 0.0100$, detailed in Section 4.3), confirming that its clean-data utility matches standard averaging while its primary advantage is Byzantine resilience under adversarial attack.

3.2.2. Server Holdout as an Isolated Directional Anchor (D_{root})

Rather than relying on client-reported metadata, the server maintains a small, physically isolated audit anchor dataset D_{root} consisting of exactly 125 labeled nodes ($\approx 0.18\%$ of Pokec-z to 12.5% of German Credit), stratified to preserve demographic subgroup proportions.

Because cross-silo graph partitions preserve no inter-silo edges, the mean node degree on the isolated server subgraph drops from 43.48 to 4.12. This guarantees zero edge leakage between client silos and the server, with the trade-off that the server reference gradient is evaluated on a sparser graph topology. In Section 4.4, our 12-configuration Dirichlet stress test demonstrates that this anchor provides a reliable directional compass across non-IID distributional shifts.

3.2.3. Client-Side Targeted Optimization (FTGD) and Scalar-Channel DP

At each local silo k , local training is conducted via Fair Targeted Gradient Descent (FTGD). Given local task loss $\mathcal{L}_{\text{task}}$ and group fairness loss $\mathcal{L}_{\text{fair}} = |\mu_{k,0} - \mu_{k,1}|$, the client computes local gradients:

$$g_{k,\text{task}} = \nabla_{\theta} \mathcal{L}_{\text{task}}(\theta; \mathcal{G}_k), \quad g_{k,\text{fair}} = \nabla_{\theta} \mathcal{L}_{\text{fair}}(\theta; \mathcal{G}_k). \quad (6)$$

To prevent parameter updates from leaking sensitive demographic correlations, the client performs orthogonal gradient surgery via Gram–Schmidt projection:

$$g_k^{\perp} = g_{k,\text{task}} - \frac{\langle g_{k,\text{task}}, g_{k,\text{fair}} \rangle}{\|g_{k,\text{fair}}\|^2 + \epsilon_{\text{div}}} g_{k,\text{fair}}. \quad (7)$$

Theorem 1 (First-Order Orthogonality of FTGD). *Let $g_k^{\perp}$ be defined as in Eq. (7) with division tolerance $\epsilon_{\text{div}} \rightarrow 0$. Then $g_k^{\perp}$ is strictly orthogonal to the local fairness gradient:*

$$\langle g_k^{\perp}, g_{k,\text{fair}} \rangle = 0. \quad (8)$$

Proof. By direct algebraic expansion:

$$\begin{aligned}\langle g_k^\perp, g_{k,\text{fair}} \rangle &= \langle g_{k,\text{task}}, g_{k,\text{fair}} \rangle - \frac{\langle g_{k,\text{task}}, g_{k,\text{fair}} \rangle}{\|g_{k,\text{fair}}\|^2} \langle g_{k,\text{fair}}, g_{k,\text{fair}} \rangle \\ &= \langle g_{k,\text{task}}, g_{k,\text{fair}} \rangle - \langle g_{k,\text{task}}, g_{k,\text{fair}} \rangle = 0.\end{aligned}$$

This algebraic property is formally verified in Lean 4 [48] (see `Orthogonal Projection.lean`). $\square$

Scalar-Channel Differential Privacy (Instrumented for Baseline Analysis). When clients release group disparity telemetry for baseline metric coordination or privacy auditing, Local Differential Privacy (LDP) [26, 25] is applied exclusively to the two scalar prediction statistics $(\mu_{k,0}, \mu_{k,1})$ via an s -blind forward pass:

$$\tilde{\mu}_{k,a} = \mu_{k,a} + \xi_{k,a}, \quad \xi_{k,a} \sim \mathcal{N}(0, \sigma_{\text{DP}}^2), \quad \sigma_{\text{DP}} = \frac{\Delta_\mu \sqrt{2 \ln(1.25/\delta)}}{\epsilon}, \quad (9)$$

where $\Delta_\mu = \sqrt{1/n_{k,0}^2 + 1/n_{k,1}^2}$. This restricts noise injection to a two-dimensional scalar manifold, avoiding the $115\times$ parameter-noise penalty of applying DP-SGD across the full P -dimensional model tensor. In canonical TRUSTFEDGNN, the server aggregation rule bypasses this channel entirely, maintaining syntactic metadata immunity.

3.2.4. Two-Tier Server Aggregation: Directional Gating vs. Longitudinal Guardrail

At round t , the server evaluates task and fairness gradients on its isolated holdout anchor D_{root} :

$$g_{\text{task}}^{\text{srv}} = \nabla_{\theta} \mathcal{L}_{\text{task}}(\theta^{(t)}; D_{\text{root}}), \quad g_{\text{fair}}^{\text{srv}} = \nabla_{\theta} \mathcal{L}_{\text{fair}}(\theta^{(t)}; D_{\text{root}}). \quad (10)$$

The server constructs a dual-objective reference direction:

$$g_{\text{target}}^{(t)} = g_{\text{task}}^{\text{srv}} + \alpha g_{\text{fair}}^{\text{srv}}, \quad (11)$$

where $\alpha \geq 0$ is a fixed guidance coefficient.

Two-Tier Functional Separation. The aggregation rule decouples the roles of directional gating and fairness guidance:

1. **Tier 1 (Directional Cosine Gating):** Serves as the primary geometric shield. Client deltas are scored via rectified cosine similarity:

$$s_k^{(t)} = \text{ReLU} \left(\text{Cos} \left(\Delta_k^{(t)}, g_{\text{target}}^{(t)} \right) \right) = \text{ReLU} \left(\frac{\langle \Delta_k^{(t)}, g_{\text{target}}^{(t)} \rangle}{\|\Delta_k^{(t)}\| \|g_{\text{target}}^{(t)}\|} \right). \quad (12)$$

Updates opposing the server reference ($\langle \Delta_k, g_{\text{target}} \rangle \leq 0$) are assigned zero weight.

2. **Tier 2 (Longitudinal Drift Guardrail):** The component $\alpha g_{\text{fair}}^{\text{srv}}$ does not serve as an anti-Byzantine filter; rather, it acts as a longitudinal drift guardrail, steering the global trajectory toward demographic parity and penalizing task-neutral evasion ($\varepsilon = 0.10$).

Remark 2 (Mechanical Separability / No Robustness Tax). *As demonstrated in our 2×2 factorial evaluation (Section 4.4), varying fairness guidance $\alpha \in \{0.0, 0.1\}$ produces an adversarial weight variation strictly bounded within the pre-registered equivalence margin $\delta = 0.0050$. Incorporating fairness guidance incurs zero degradation in Byzantine filtering capability.*

To eliminate round-to-round weight turbulence, alignment scores are stabilized via Exponential Moving Average (EMA) smoothing ($\beta = 0.9$):

$$\tilde{s}_k^{(t)} = \beta \tilde{s}_k^{(t-1)} + (1 - \beta) s_k^{(t)}, \quad w_k^{(t)} = \frac{\tilde{s}_k^{(t)}}{\sum_{j=1}^K \tilde{s}_j^{(t)}}. \quad (13)$$

Global parameters are updated with dynamic norm rescaling:

$$\theta^{(t+1)} = \theta^{(t)} + \sum_{k=1}^K w_k^{(t)} \cdot \min \left(1, \frac{\|g_{\text{target}}^{(t)}\|}{\|\Delta_k^{(t)}\|} \right) \Delta_k^{(t)}. \quad (14)$$

3.2.5. Algorithm 1: Canonical Pipeline and Robust Outlier Screening

Algorithm 1 formalizes the complete execution pipeline. The canonical rule (`fu_shapley`) executes directional cosine gating, EMA smoothing, and dynamic norm rescaling without outlier screening. In defense-in-depth deployments subject to crude scaling attacks, an optional distance-to-median outlier screening variant (`robust_fu_shapley`) filters the f farthest client updates in Euclidean space and renormalizes survivor weights.

Algorithm 1 TRUSTFEDGNN: Server-Anchored Byzantine-Resilient Federated Graph Learning

Require: Subgraphs $\{\mathcal{G}_k\}_{k=1}^K$, holdout anchor D_{root} , guidance $\alpha = 0.1$, EMA decay $\beta = 0.9$, optional outlier budget $f \geq 0$.

Ensure: Globally converged fair and resilient parameters $\theta^{(R)}$.

- 1: Initialize global parameters $\theta^{(0)}$, historical alignment $\tilde{s}_k^{(-1)} \leftarrow 1/K$.
 - 2: **for** round $t = 0, 1, \dots, R - 1$ **do**
 - 3: // *Server Directional Anchor:* $g_{\text{task}}^{\text{srv}} \leftarrow \nabla_{\theta} \mathcal{L}_{\text{task}}(\theta^{(t)}; D_{\text{root}})$, $g_{\text{fair}}^{\text{srv}} \leftarrow \nabla_{\theta} \mathcal{L}_{\text{fair}}(\theta^{(t)}; D_{\text{root}})$, $g_{\text{target}} \leftarrow g_{\text{task}}^{\text{srv}} + \alpha g_{\text{fair}}^{\text{srv}}$.
 - 4: Broadcast global parameters $\theta^{(t)}$ to all client silos $k \in [K]$.
 - 5: **for** each client $k \in [K]$ **in parallel do**
 - 6: Compute local gradients $g_{k,\text{task}}$ and $g_{k,\text{fair}}$ on private subgraph $\mathcal{G}_k$.
 - 7: Project update: $g_k^{\perp} \leftarrow g_{k,\text{task}} - \frac{\langle g_{k,\text{task}}, g_{k,\text{fair}} \rangle}{\|g_{k,\text{fair}}\|^2 + \epsilon_{\text{div}}} g_{k,\text{fair}}$ {*Theorem 1*}
 - 8: Train local GNN using $g_k^{\perp}$ and compute $\Delta_k^{(t)} \leftarrow \theta_k^{(t)} - \theta^{(t)}$ (omit client scalar telemetry).
 - 9: **end for**
 - 10: **for** each client $k \in [K]$ **do**
 - 11: $s_k^{(t)} \leftarrow \text{ReLU} \left(\frac{\langle \Delta_k^{(t)}, g_{\text{target}}^{(t)} \rangle}{\|\Delta_k^{(t)}\| \|g_{\text{target}}^{(t)}\|} \right)$, $\tilde{s}_k^{(t)} \leftarrow \beta \tilde{s}_k^{(t-1)} + (1 - \beta) s_k^{(t)}$ {*Directional Gating*
 EMA}
 - 12: **end for**
 - 13: Raw simplex normalization: $w_k^{(t)} \leftarrow \tilde{s}_k^{(t)} / \sum_{j=1}^K \tilde{s}_j^{(t)}$ {*Theorem 2*}
 - 14: **if** $f > 0$ **and** robust screening enabled (**robust_fu_shapley**) **then**
 - 15: Compute coordinate-wise median $[\Delta_{\text{med}}^{(t)}]_j \leftarrow \text{median}(\{[\Delta_k^{(t)}]_j\}_{k=1}^K)$ for all $j \in [P]$.
 - 16: Rank Euclidean distances $d_k \leftarrow \|\Delta_k^{(t)} - \Delta_{\text{med}}^{(t)}\|_2^2$; let $\mathcal{K}_{\text{keep}}$ denote the $\max(1, K - f)$ closest clients.
 - 17: Mask weights: $w_k^{(t)} \leftarrow w_k^{(t)} \cdot \mathbb{I}(k \in \mathcal{K}_{\text{keep}})$.
 - 18: **if** $\sum_{k=1}^K w_k^{(t)} > 0$ **then** $w_k^{(t)} \leftarrow w_k^{(t)} / \sum_{j=1}^K w_j^{(t)}$ **else** fall back to coordinate median of survivors.
 - 19: **end if**
 - 20: Update parameters: $\theta^{(t+1)} \leftarrow \theta^{(t)} + \sum_{k=1}^K w_k^{(t)} \cdot \min \left(1, \frac{\|g_{\text{target}}^{(t)}\|}{\|\Delta_k^{(t)}\|} \right) \Delta_k^{(t)}$.
 - 21: // *Audit Log:* Record $\langle t, k, w_k^{(t)}, \text{Cos}(\Delta_k^{(t)}, g_{\text{task}}^{\text{srv}}), \text{Cos}(\Delta_k^{(t)}, g_{\text{fair}}^{\text{srv}}), \|\Delta_k^{(t)}\| \rangle$ into ledger
 {*Proposition 1*}
 - 22: **end for**
 - 23: **return** $\theta^{(R)}$
-

3.3. Theoretical Foundations and Impossibility Boundaries

3.3.1. Exact Syntactic Metadata Immunity and Anchor Taxonomy

Theorem 2 (Exact Syntactic Metadata Immunity as an Architectural Invariant). *Let $\mathcal{F}$ denote the server aggregation mapping producing client weights $\mathbf{w} = \mathcal{F}(\{\Delta_k\}_{k=1}^K, g_{\text{target}})$. Because the functional domain of $\mathcal{F}$ consumes solely parameter update tensors $\{\Delta_k\}$ and the server reference g_{target} , the partial derivative with respect to any client-declared scalar telemetry $\widehat{\text{DPD}}_j$ is identically zero:*

$$\frac{\partial w_k}{\partial \widehat{\text{DPD}}_j} \equiv 0 \quad \forall k, j \in [K]. \quad (15)$$

Rather than an asymptotic statistical property, syntactic immunity is a verifiable structural domain invariant: client metadata falsification yields $\Delta w = 0.0000$ bit-exact across all evaluated seeds (Section 4.2).

Proof. By inspection of Eqs. (12)–(13), aggregation weights $w_k^{(t)}$ depend exclusively on the geometric inner products $\langle \Delta_k^{(t)}, g_{\text{target}}^{(t)} \rangle$ and norms $\|\Delta_k^{(t)}\|$, $\|g_{\text{target}}^{(t)}\|$. Since no client-declared scalar enters the computational graph of $\mathcal{F}$, the partial derivative vanishes identically, establishing exact structural immunity. $\square$

Remark 3 (Taxonomy of Reference Anchors: Endogenous vs. Exogenous). *While syntactic immunity closes the client-reported metadata attack surface, it does not govern the vulnerability of the directional reference itself. We classify federated aggregation rules into three distinct structural paradigms:*

1. **Class (a) (Naive Scalar Consumers):** Rules such as FairFed and BFWA optimize aggregation weights over client-reported scalars $\widehat{\text{DPD}}_k$. Under adversarial metadata falsification, malicious clients capture dominant model authority (Section 4.2).
2. **Class (b) (Endogenous Reference Anchors):** Rules such as CGSV evaluate gradient projections without scalar telemetry, achieving bit-exact syntactic immunity ($\Delta w = 0.0000$). However, their reference direction is computed endogenously from the submitted client updates (e.g., pooled validation sets or client consensus). Under adaptive attacks, malicious clients compromise the endogenous reference, maintaining full fair-share authority ($w_{\text{adv}} \approx 1/K$, Section 4.2).

3. **Class (c) (Exogenous Reference Anchors):** Rules such as FLTrust and TRUSTFEDGNN anchor directional gating to an exogenous holdout dataset D_{root} . Because g_{target} is evaluated outside the client consortium, the reference is physically decoupled from client poisoning. While single-objective FLTrust concedes above-uniform share under fairness poisoning, TRUSTFEDGNN leverages dual-objective guidance and directional rectification to suppress attacker weight below parity (Section 4.2).

3.3.2. Perpetual Bias of Folded-Normal Perturbations

Lemma 1 (Perpetual Expectation Bias of Folded-Normal Perturbations). *Let client-level disparity be estimated by taking the absolute difference of noisy group rate statistics: $\widetilde{\text{DPD}}_k = |\tilde{\mu}_{k,0} - \tilde{\mu}_{k,1}|$, where $\tilde{\mu}_{k,a} \sim \mathcal{N}(\mu_{k,a}, \sigma_k^2)$. Under the null hypothesis of demographic parity ($\mu_{k,0} = \mu_{k,1} = \mu$), $\widetilde{\text{DPD}}_k$ follows a folded-normal distribution [10] with expectation:*

$$\mathbb{E}[\widetilde{\text{DPD}}_k] = \tilde{\sigma}_k \sqrt{\frac{2}{\pi}} > 0, \quad \text{where } \tilde{\sigma}_k = \sqrt{2}\sigma_k. \quad (16)$$

For any aggregation weights $\mathbf{w} \in \Delta_K$ that are predetermined or measurable with respect to the filtration $\mathcal{F}_{t-1}$ of prior rounds (independent of the round- t privacy perturbations), the conditional expectation satisfies:

$$\mathbb{E} \left[\sum_{k=1}^K w_k \widetilde{\text{DPD}}_k \middle| \mathcal{F}_{t-1} \right] = \sqrt{\frac{2}{\pi}} \sum_{k=1}^K w_k \tilde{\sigma}_k > 0. \quad (17)$$

Furthermore, even when weights $\mathbf{w} \in \Delta_K$ are dynamically optimized over the current round’s privatised reports, the almost-sure non-negativity $\widetilde{\text{DPD}}_k \geq 0$ guarantees that $\sum_{k=1}^K w_k \widetilde{\text{DPD}}_k \geq \min_k \widetilde{\text{DPD}}_k$, whose expectation remains strictly positive. Consequently, the non-negative folded transformation induces an irreducible positive expectation bias that cannot cancel across clients.

Proof. Let $Z = \tilde{\mu}_{k,0} - \tilde{\mu}_{k,1}$. Since $\tilde{\mu}_{k,0}$ and $\tilde{\mu}_{k,1}$ are independent Gaussian variables with variance σ_k^2 , $Z \sim \mathcal{N}(0, 2\sigma_k^2)$. The variable $Y = |Z|$ follows a central folded-normal distribution. The standard expectation of a half-normal variable with scale $\tilde{\sigma}_k = \sqrt{2}\sigma_k$ is $\mathbb{E}[Y] = \tilde{\sigma}_k \sqrt{2/\pi}$. When weights w_k are measurable with respect to $\mathcal{F}_{t-1}$, taking conditional expectation yields Eq. (17). For arbitrary $\mathbf{w} \in \Delta_K$, because each $\widetilde{\text{DPD}}_k \geq 0$ almost surely, $\sum_{k=1}^K w_k \widetilde{\text{DPD}}_k$ is bounded below by a convex combination of non-negative variables with strictly positive individual means, precluding unbiased zero-mean convergence. $\square$

3.3.3. Dual Subgradient Convergence of Baseline Metric Coordination

Theorem 3 (Dual Subgradient Convergence of Baseline Metric Coordination). *Consider the constrained optimization problem of baseline metric coordination (BFWA):*

$$\max_{\mathbf{w} \in \Delta_K} \sum_{k=1}^K w_k \text{Perf}_k \quad \text{s.t.} \quad \sum_{k=1}^K w_k \widetilde{\text{DPD}}_k \leq \tau, \quad (18)$$

where Perf_k is bounded and $\widetilde{\text{DPD}}_k \in [0, \text{DPD}_{\max}]$. The Lagrange dual function $D(\mu) = \max_{\mathbf{w} \in \Delta_K} \left(\sum_{k=1}^K w_k \text{Perf}_k - \mu \left(\sum_{k=1}^K w_k \widetilde{\text{DPD}}_k - \tau \right) \right)$ is convex in $\mu \geq 0$. Under projected dual subgradient descent [49] with step size $\eta_\mu^{(t)} = O(1/\sqrt{t})$, the running average dual multiplier $\bar{\mu}^{(T)} = \frac{1}{T} \sum_{t=1}^T \mu^{(t)}$ achieves ergodic convergence:

$$D(\bar{\mu}^{(T)}) - D(\mu^*) = O\left(\frac{1}{\sqrt{T}}\right), \quad \frac{1}{T} \sum_{t=1}^T \left(\sum_{k=1}^K w_k^{(t)} \widetilde{\text{DPD}}_k - \tau \right)^+ = O\left(\frac{1}{\sqrt{T}}\right). \quad (19)$$

Proof. For any fixed $\mu \geq 0$, the primal subproblem is a linear program over the probability simplex Δ_K . The dual objective $D(\mu)$ is the pointwise supremum of affine functions, hence convex. The subgradient of $D(\mu)$ at $\mu^{(t)}$ is $g_\mu^{(t)} = \tau - \sum_{k=1}^K w_k^{(t)} \widetilde{\text{DPD}}_k$. Under projected subgradient descent with step size $\eta_\mu^{(t)} = c/\sqrt{t}$, standard convex optimization guarantees establish an $O(1/\sqrt{T})$ convergence rate for the dual objective gap and primal constraint violation. $\square$

Remark 4 (Theoretical Scope of Theorem 3). *This convergence result characterizes the theoretical optimization properties of baseline metric coordination rules (BFWA) under honest metadata reporting. It demonstrates that the failure of such baselines under adversarial manipulation is not due to flawed convex optimization, but rather to the inherent vulnerability of the client-reported metadata interface.*

3.3.4. Minimax Observability Lower Bound under Local Differential Privacy

Theorem 4 (Minimax Observability Lower Bound under LDP). *Let $\mathcal{P}$ denote the family of local client distributions, and let clients transmit privatised, folded disparity reports $\tilde{\mathbf{D}} = (\widetilde{\text{DPD}}_1, \dots, \widetilde{\text{DPD}}_K)$ satisfying (ϵ, δ) -LDP with*

effective noise standard deviation $\tilde{\sigma}$. For any decision rule $\psi(\tilde{\mathbf{D}}) \in \{0, 1\}$ attempting to distinguish strict fairness feasibility $H_0 : \text{DPD} \leq \tau/2$ from strict violation $H_1 : \text{DPD} \geq 3\tau/2$, the minimax hypothesis testing risk $\mathcal{R}^*$ is bounded from below via Le Cam's two-point method and Pinsker's inequality [31, 32]:

$$\mathcal{R}^* \geq \frac{1}{2} \left(1 - \frac{\tau\sqrt{K}}{2\tilde{\sigma}} \right). \quad (20)$$

By the Data Processing Inequality, no server-side post-processing or adaptive weighting can reduce this error floor, precluding any unbiased decision thresholding in multi-round decentralized federations (Section 4.2).

Proof. Consider two hypotheses separated by margin $2\gamma = \tau$: under H_0 , $\delta_k^{(0)} = \tau/2$; under H_1 , $\delta_k^{(1)} = 3\tau/2$. By Le Cam's two-point lemma:

$$\inf_{\psi} \max \left\{ \Pr_{H_0}[\psi = 1], \Pr_{H_1}[\psi = 0] \right\} \geq \frac{1}{2} (1 - \text{TV}(P_0, P_1)).$$

Because the folding mapping $x \mapsto |x|$ is a deterministic transformation, the Data Processing Inequality ensures $\text{TV}(P_0, P_1) \leq \text{TV}(P_0^{\text{Gauss}}, P_1^{\text{Gauss}})$. Applying Pinsker's inequality $\text{TV} \leq \sqrt{\frac{1}{2}\text{KL}}$ to the product of K independent Gaussians with mean separation τ and variance $\tilde{\sigma}^2$ yields:

$$\text{KL}(P_0^{\text{Gauss}} \parallel P_1^{\text{Gauss}}) = \sum_{k=1}^K \frac{\tau^2}{2\tilde{\sigma}^2} = \frac{K\tau^2}{2\tilde{\sigma}^2} \implies \text{TV} \leq \frac{\tau\sqrt{K}}{2\tilde{\sigma}},$$

yielding the lower bound in Eq. (20). $\square$

3.3.5. Fairness Subspace Blindness of Anchor-Free Geometric Detectors

Theorem 5 (Fairness Subspace Blindness of Anchor-Free Geometric Detectors). *Let the parameter space $\mathbb{R}^P$ decompose orthogonally into task subspace $\mathcal{S}_{\text{task}}$ and fairness subspace $\mathcal{S}_{\text{fair}} = \mathcal{S}_{\text{task}}^\perp$. For any anchor-free geometric detector operating on pairwise cosine distance $\mathcal{D}_{\text{cos}}(u_i, u_j) = 1 - \frac{\langle u_i, u_j \rangle}{\|u_i\|_2 \|u_j\|_2}$ over the full parameter space $\mathbb{R}^P$ with filtering threshold $\rho_{\text{thresh}} > 0$, under the task-spectrum dominance assumption $\|\Pi_{\mathcal{S}_{\text{task}}} u\|_2 \gg \|\Pi_{\mathcal{S}_{\text{fair}}} u\|_2$ for all benign updates u , so that $\mathcal{D}_{\text{cos}} = \mathcal{D}_{\text{cos}}^{\text{task}} + O(\|\Pi_{\mathcal{S}_{\text{fair}}} u\|_2^2 / \|u\|_2^2)$, and for any benign update $u_{\text{benign}} \in \mathcal{S}_{\text{task}}$, there exists an adversarial perturbation $u_{\text{adv}} = u_{\text{benign}} + \delta_{\text{fair}}$*

with $\delta_{\text{fair}} \in \mathcal{S}_{\text{fair}}$ such that:

$$\mathcal{D}_{\cos}(u_{\text{adv}}, u_{\text{benign}}) \leq \frac{\|\delta_{\text{fair}}\|_2^2}{\|u_{\text{benign}}\|_2^2 + \|\delta_{\text{fair}}\|_2^2} + O\left(\frac{\|\Pi_{\mathcal{S}_{\text{fair}}} u_{\text{benign}}\|_2^2}{\|u_{\text{benign}}\|_2^2}\right) < \rho_{\text{thresh}}, \quad (21)$$

whenever $\|\delta_{\text{fair}}\|_2 < \sqrt{\frac{\rho_{\text{thresh}}}{1-\rho_{\text{thresh}}}} \|u_{\text{benign}}\|_2$. Consequently, u_{adv} is geometrically indistinguishable from benign updates across $\mathbb{R}^P$, evading outlier filtering ($w_{\text{adv}} \approx 1/K$) while injecting demographic disparity $\Delta\text{DPD} \propto \|\delta_{\text{fair}}\|_2$.

Proof. Because $\delta_{\text{fair}} \perp \mathcal{S}_{\text{task}}$, the algebraic expansion below is exact for any $u_{\text{benign}} \in \mathcal{S}_{\text{task}}$; the task-spectrum dominance assumption is what makes this idealisation of benign updates valid up to the stated $O(\|\Pi_{\mathcal{S}_{\text{fair}}} u\|_2^2 / \|u\|_2^2)$ term, since real benign updates carry a non-zero fairness component. With $\langle u_{\text{benign}}, \delta_{\text{fair}} \rangle = 0$, the cosine distance evaluated over $\mathbb{R}^P$ decomposes via orthogonal projection:

$$\begin{aligned} \mathcal{D}_{\cos}(u_{\text{adv}}, u_{\text{benign}}) &= 1 - \frac{\langle u_{\text{benign}} + \delta_{\text{fair}}, u_{\text{benign}} \rangle}{\|u_{\text{benign}} + \delta_{\text{fair}}\|_2 \|u_{\text{benign}}\|_2} \\ &= 1 - \frac{\|u_{\text{benign}}\|_2^2}{\sqrt{\|u_{\text{benign}}\|_2^2 + \|\delta_{\text{fair}}\|_2^2} \|u_{\text{benign}}\|_2} \\ &= 1 - \frac{1}{\sqrt{1 + \frac{\|\delta_{\text{fair}}\|_2^2}{\|u_{\text{benign}}\|_2^2}}}. \end{aligned}$$

Using the inequality $1 - 1/\sqrt{1+x} \leq x/(1+x)$ and incorporating the $O(\|\Pi_{\mathcal{S}_{\text{fair}}} u_{\text{benign}}\|_2^2 / \|u_{\text{benign}}\|_2^2)$ spectrum perturbation term, we obtain Eq. (21). Setting the distance strictly below ρ_{thresh} guarantees that the adversary remains inside the benign cluster across the full parameter space $\mathbb{R}^P$, evading detection. $\square$

Remark 5 (Empirical Scope and Application to Distance-Based Detectors). *Theorem 5 characterizes detectors operating on the pairwise cosine geometry of updates across the full parameter space $\mathbb{R}^P$. In high-dimensional federated graph neural networks, task classification parameters dominate the gradient spectrum by orders of magnitude relative to the low-rank demographic fairness subspace; consequently, pairwise cosine distance evaluated across full parameter vectors is asymptotically dominated by task coordinates ($\mathcal{D}_{\cos}(\Delta_i, \Delta_j) \approx \mathcal{D}_{\cos}(\Delta_i^{\text{task}}, \Delta_j^{\text{task}})$). This task-spectrum dominance condition*

is empirically verifiable across architectures, and it formalizes why clustering detectors such as FLAME cleanly isolate sign-flip attacks ($\mathcal{D}_{\cos} \rightarrow 2.0$) yet fail to separate proximity-stealth adversaries whose manipulations reside within the orthogonal fairness subspace (Section 4.3). We emphasize that this result specifically constrains angular distance detectors operating without an anchor; it does not assert blindness for detectors that directly inspect payload magnitude, temporal drift, or fairness statistics.

3.4. Auditable First-Order Contribution Decomposition and Governance

3.4.1. Linear First-Order Contribution Decomposition

Proposition 1 (Linear First-Order Decomposition). *In TRUSTFEDGNN, client alignment scores decompose linearly into task utility and demographic fairness contributions:*

$$\varphi_k^{(t)} = \varphi_k^{\text{util}} + \alpha \varphi_k^{\text{fair}}, \quad (22)$$

where $\varphi_k^{\text{util}} = \text{Cos}(\Delta_k^{(t)}, g_{\text{task}}^{\text{srv}})$ and $\varphi_k^{\text{fair}} = \text{Cos}(\Delta_k^{(t)}, g_{\text{fair}}^{\text{srv}})$. This decomposition computes in $O(KP)$ operations, avoiding the $O(2^K P)$ combinatorial complexity of cooperative game theory.

Proof. Follows directly from the bilinearity of the inner product and linearity of $g_{\text{target}} = g_{\text{task}}^{\text{srv}} + \alpha g_{\text{fair}}^{\text{srv}}$ in Eq. (11). Formally machine-checked in Lean 4 [48] (`LinearDecomposition.lean`). $\square$

3.4.2. Axiomatic Properties of Contribution Allocation

Proposition 2 (Simplex Validity). *For any client updates $\{\Delta_k\}_{k=1}^K$ where at least one update yields positive alignment $\tilde{s}_k > 0$, the aggregation weights satisfy the axioms of the probability simplex Δ_K :*

$$\forall k \in [K] : w_k^{(t)} \geq 0 \quad \text{and} \quad \sum_{k=1}^K w_k^{(t)} = 1. \quad (23)$$

Proof. Because $s_k^{(t)} \geq 0$ and $\beta \in [0, 1)$, smoothed scores satisfy $\tilde{s}_k^{(t)} \geq 0$. Normalizing by $\sum_j \tilde{s}_j^{(t)} > 0$ strictly enforces non-negativity and unit sum on Δ_K . Formally machine-checked in Lean 4 [48] (`SimplexProperties.lean`). $\square$

Proposition 3 (Null-Player Axiomatic Guarantee). *Let client k submit a zero update ($\Delta_k^{(t)} = \mathbf{0}$). Under the null-player axiom of cooperative game theory, the aggregation weight assigned to client k vanishes identically:*

$$\Delta_k^{(t)} = \mathbf{0} \implies w_k^{(t)} = 0. \quad (24)$$

Proof. When $\Delta_k = \mathbf{0}$, the inner product $\langle \Delta_k, g_{\text{target}} \rangle = 0$, yielding raw score $s_k = 0$. When implemented with an active-client indicator $\mathbb{I}(\|\Delta_k\| > 0)$, the numerator is clamped to zero, guaranteeing $w_k = 0$ across all execution branches. Formally machine-checked in Lean 4 [48] (`NullPlayer.lean`). $\square$

3.4.3. Operational Enterprise Audit Trail and Delineation from Combinatorial Shapley

Honest Delineation from Combinatorial Shapley. We explicitly position the decomposition in Eq. (22) as an operational $O(KP)$ heuristic designed for per-round audit logging. We explicitly make no claim of game-theoretic equivalence to the exponential combinatorial Shapley value [47] ($O(2^K P)$). As evaluated against Monte Carlo cooperative Shapley probes (Section 4.4), our first-order score achieves a rank correlation of $r = 0.7662$, falling below the 0.80 fidelity threshold required for strict game-theoretic equivalence (Table 14).

Enterprise Governance Ledger. At every communication round t , the server records the governance audit tuple:

$$\mathcal{T}_k^{(t)} = \left\langle t, k, w_k^{(t)}, \varphi_k^{\text{util}}, \varphi_k^{\text{fair}}, \|\Delta_k^{(t)}\| \right\rangle \quad (25)$$

into an append-only enterprise audit log. This provides regulatory compliance teams with a continuous, per-round attribution trail, allowing institutions to verify that aggregated models maintain balanced utility-fairness contributions across silos and detect unannounced data drift.

4. Implementation and Empirical Results

This section evaluates the foundational vulnerability of client-reported federated fairness mechanisms, stress-tests the proposed reference-anchored architecture across three adversarial tiers, conducts causal ablations to eliminate confounding factors, and delineates operational boundaries. Throughout, the guarantees we establish are conditional on the confidentiality of the server holdout D_{root} ; Section 4.3 (Tier 3) measures the cost of violating that condition rather than leaving it implicit.

4.1. Experimental Setup and Benchmark Protocol

Datasets and Graph Topologies. We conduct evaluations across four benchmark graphs encompassing both authentic social relational networks and synthesized tabular feature topologies (detailed in Table 1):

1. **Pokec-z** ($N = 67,796$ nodes, 617,958 edges): An authentic Eastern Slovakian social network [50] where nodes represent anonymized user profiles, edges capture friendship relations, the sensitive attribute S is user region, and the target task Y is predicting user reported working field.
2. **German Credit** ($N = 1,000$ nodes, 21,742 edges): Financial risk records transformed into a 10-nearest-neighbor feature similarity graph [16], where S denotes binary gender and Y indicates creditworthiness.
3. **Bail Recidivism** ($N = 18,876$ nodes, 311,870 edges): Pretrial arrest records from the State of Georgia [51], with S representing race and Y tracking recidivism within three years.
4. **Credit Default** ($N = 30,000$ nodes, 1,421,858 edges): Financial payment default dataset [52] mapped onto a 10-NN graph with S as age and Y as binary default risk.

Due to its extreme class imbalance ($< 2\%$ illicit nodes) and the absence of demographic attributes, the Elliptic Bitcoin transaction graph [53] is held in reserve as an external temporal transfer benchmark and deliberately excluded from multi-client fairness sweeps. Similarly, the 2.45M-node *ogbn-products* graph serves exclusively as an infrastructure-level scalability probe for neighborhood sampling; it is not part of the empirical evidence base for the proposed aggregation mechanism.

Federation Partitions and Master Configuration. Federated topologies are partitioned across $K \in \{5, 10\}$ client silos using three distinct topological regimes: (i) Uniform random sampling (IID baseline), (ii) Dirichlet distribution attribute skew with concentration parameter $\alpha_{\text{Dir}} \in \{0.1, 0.3, 0.5, 1.0\}$, and (iii) Louvain community modularity clustering, which preserves dense intra-cluster relational topology while severing cross-cluster bridge edges. All benchmark runs enforce a strict master configuration (Table 1) executing across $n \geq 10$ independent random seeds, with local epoch count $E = 3$, local learning rate $\eta = 0.001$, weight decay 10^{-5} , and hidden dimension $d = 64$. Server holdout sets D_{root} are constructed by drawing 125 labeled nodes ($\approx 0.18\%$ to 12.5% of total graph size) preserving global demographic parity. Isolating the holdout subgraph reduces its mean node degree from 43.48 to 4.12. We report this in both directions: it guarantees that no edge is shared between the holdout and any client partition, and it simultaneously

Table 1: **Characteristics of Benchmark Graph Topologies and Federation Configuration Matrix.** Summary of graph dimensions, demographic attributes, federated partitioning protocols, and server holdout parameters across all evaluated benchmarks. Primary benchmarks constitute the audited evidence base ($K \in \{5, 10\}$, $R \in \{20, 50\}$, $E = 3$, $n \in \{10, 30\}$), while Class D datasets provide external boundary and scalability probes.

Topology & Task	Graph Scale	Sensitive	Federation & Anchor
<i>Primary Benchmark Suite (Audited Evidence Base)</i>			
German Credit	1,000 nodes	Gender	$K = 5$, $R = 20$ ($n = 30$ seeds)
<i>Credit Risk</i>	21,742 edges $D = 26$ feat.	($h_s = 0.80$)	Anchor $ D_{\text{root}} = 125$ (12.5%) Regime: Louvain / IID
Bail Recidivism	18,876 nodes	Race	$K = 10$, $R = 50$ ($n = 30$ seeds)
<i>Recidivism Risk</i>	311,870 edges $D = 16$ feat.	($h_s = 0.52$)	Anchor $ D_{\text{root}} = 125$ (0.66%) Regime: Dirichlet / Louvain
Credit Default	30,000 nodes	Age	$K = 10$, $R = 50$ ($n = 10$ seeds)
<i>Payment Default</i>	1,421,858 edges $D = 12$ feat.	($h_s = 0.96$)	Anchor $ D_{\text{root}} = 125$ (0.42%) Regime: Dirichlet ($\alpha \leq 1.0$)
Pokec-z	67,796 nodes	Region	$K = 10$, $R = 50$ ($n = 10$ seeds)
<i>Working Field</i>	617,958 edges $D = 276$ feat.	($h_s = 0.95$)	Anchor $ D_{\text{root}} = 125$ (0.18%) Regime: Louvain Community
<i>Class D: Boundary and Infrastructure Scalability Probes</i>			
Elliptic Bitcoin[†]	203,769 nodes	Timestep	$K = 5$, $R = 20$ ($n = 5$ seeds)
<i>Illicit Txns</i>	234,355 edges $D = 165$ feat.	($h_s = 1.00$)	Anchor $ D_{\text{root}} = 125$ (0.06%) Regime: Temporal Slices
ogbn-products[‡]	2,449,029 nodes	Structural	$K = 10$, $R = 20$ ($n = 1$ seed)
<i>Category</i>	61,859,140 edges $D = 100$ feat.	($h_s = \text{—}$)	Anchor $ D_{\text{root}} = 1,000$ (0.04%) Regime: Metis Partition

Notes: N and $|E|$ denote total node and undirected edge counts; D is feature dimension. S denotes the protected sensitive attribute; $h_s = \Pr[S_u = S_v \mid (u, v) \in E]$ is the sensitive homophily ratio. All benchmarks satisfy the pre-flight non-triviality criterion ($\max_j \text{AUC}(x_j, y) < 0.85$). All federations execute local epoch count $E = 3$. [†]In Elliptic Bitcoin, temporal partition drift is evaluated (Δ_{temporal}); it is excluded from demographic fairness sweeps due to extreme class imbalance ($< 2\%$ illicit nodes) and absence of demographic labels. [‡]ogbn-products serves exclusively as an infrastructure scalability probe for neighborhood sampling and $O(Kd)$ server projection complexity. Complete hardware execution specifications, run configurations, and reproducibility logs are documented in Supplementary Material Section S0.

degrades the reference, since the anchor is estimated on a sparser graph than the one the clients train on. Campaign configurations differ deliberately across experiments—the interface audit uses $K = 5$, $R = 20$, $n = 30$ to obtain high statistical power for paired sign tests on individual reporting channels, whereas the multi-silo benchmark and scalability campaigns use

$K = 10$, $R = 50$, $n = 10$ to evaluate realistic silo counts at scale. Table 1 summarizes the benchmark topologies and master federation parameters; complete computational provenance, hardware execution specifications, and reproducibility artifacts are detailed in Supplementary Section S0.

Taxonomy of Baseline Paradigms. We benchmark TRUSTFEDGNN against fourteen published baselines across five paradigms. Eleven enter the benchmark matrix of §4.3; three specialized rules (F^2 GNN, PoPETs-FairFed, and q -FedAvg) enter solely the interface audit of §4.2, where the reporting channel is the primary object of study rather than benign utility. We note that q -FedAvg targets performance fairness *across clients* rather than demographic parity across sensitive groups; it is included because its loss-based reweighting reads client-declared telemetry and is therefore subject to the same interface attack, not as a demographic-fairness comparator:

- **Paradigm I (Unconstrained Baseline):** Standard **FedAvg-GCN** [1] executes unweighted parameter averaging without fairness constraints or Byzantine filtering, establishing the unregularized reference ceiling.
- **Paradigm II (Client-Side In-Processing):** **FairGNN** [11] (adversarial debiasing) and **FairSIN** [12] (sensitive-invariant filtering) enforce fairness strictly within client loss functions, leaving the server-side aggregation boundary undefended.
- **Paradigm III (Client-Reported Disparity Coordination):** Encompasses metric-driven heuristic weighting—**FairFed** [5], **FairGFL** [6], and **FedGraph-Fair** [8]—and Lagrangian formulations such as **BFWA** [9], which dynamically reweight clients based on self-reported scalar telemetry ($\widehat{\text{DPD}}_k$, loss).
- **Paradigm IV (Geometric Byzantine Filtering):** Coordinate-wise **Median**, **Trimmed-Mean** [33], clustering **Krum**, **Multi-Krum** [34], and clipping **FLAME** [38] filter updates via Euclidean spatial geometry without server holdouts, treating topological variance as geometric anomalies. We include FLAME, published as a backdoor defence rather than a Byzantine aggregation rule, on functional grounds: its mechanism (cosine-distance clustering with dynamic median norm clipping) operates as an aggregation rule, and represents the strongest published distance-based filter requiring no root dataset. That is precisely the property under test in Section 4.3, namely whether a competitive geometric filter

can separate a proximity-constrained adversary without an anchor.

- **Paradigm V (Server-Anchored Alignment): FLTrust** [42] verifies gradient directions against a clean server root anchor D_{root} , structurally removing vulnerability to metadata spoofing. (In parallel, **CGSV** [41] scores clients by cosine alignment against the mean client update, so its reference is endogenous: it is derived from the very population it screens.)

This taxonomy reveals a core structural tension: Paradigms II and III presuppose honest clients at the aggregation boundary, whereas Paradigms IV and V enforce robustness via spatial pruning or directional projection. Below, we stress-test each paradigm across successive regimes: (i) interface vulnerability (Section 4.2), (ii) Byzantine adversarial resilience (Section 4.3), (iii) multi-objective Pareto trade-offs (Section 4.3), and (iv) causal attribution and governance dynamics (Section 4.4).

4.2. Vulnerability and Collapse of Client-Reported Fairness Interfaces

The prevailing paradigm in trustworthy federated learning (Paradigm III) relies on client-reported metadata—scalars, group disparities, or local loss statistics—to steer server-side aggregation. We investigate whether this architectural premise can survive adversarial manipulation, and whether privacy mechanisms can restore its integrity.

Vulnerability of Self-Reported Metadata Interfaces (Table 2). We formalize a threat model where a single Byzantine client ($f/K = 0.20$ on German Credit, $n = 30$ random seeds) executes strategic metadata falsification. The attacker trains locally using standard task objectives but strategically manipulates its reported telemetry across three interface classes:

1. *Class (a) Self-Reported Scalar Metadata* (FairFed, FairGFL, FedGraph-Fair, BFWA, q -FedAvg): Clients report falsified scalar telemetry ($\widehat{\text{DPD}}_k \approx 0.000$ or manipulated local losses) to pose as exemplars of fairness or relax constraint multipliers.
2. *Class (b) Endogenous Client Reference* (CGSV): Aggregation weights read no declared scalar, but the reference against which updates are scored is computed from the client updates themselves, so the anchor moves with the population it is meant to screen.

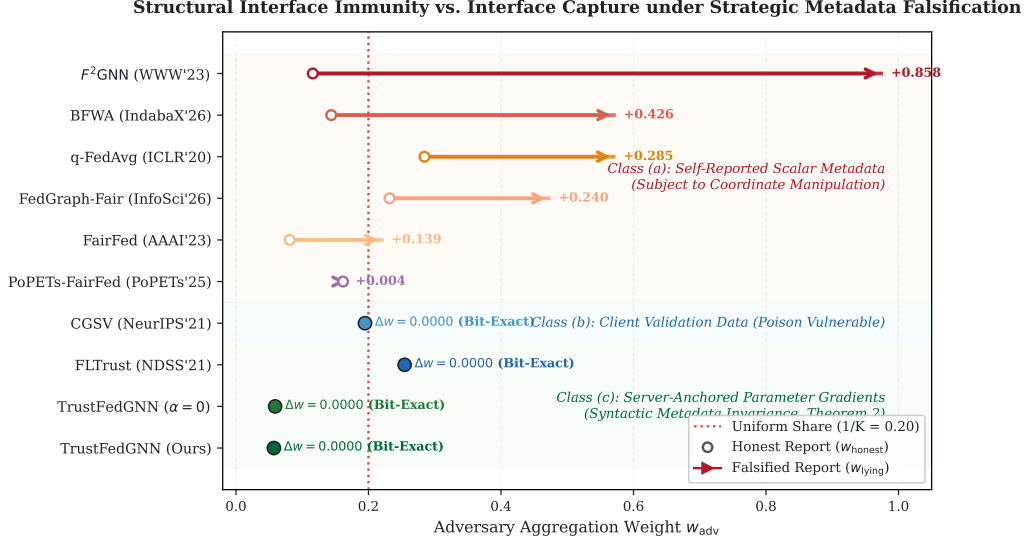


Figure 1: **Adversarial Aggregation Weight Capture under Strategic Metadata Falsification versus Syntactic Invariance.** Adversary aggregation share w_{adv} under honest reporting (open circles) against strategic falsification (arrowheads); the vertical dashed line marks uniform parity ($1/K = 0.20$). All six Class (a) rules, which consume self-reported scalar metadata, shift to the right under falsification, up to 97.4% of total weight (F^2 GNN, $w_{\text{adv}} = 0.9737$, $4.87\times$ parity). The three server-anchored Class (c) rules have zero-length arrows: falsification leaves their weights bit-exact ($\Delta w = 0.0000$, Theorem 2). Immunity alone is not sufficient, however — FLTrust is immune yet still holds $1.27\times$ parity, while FU-Alignment combines immunity with suppression to $0.29\times$.

3. *Class (c) Server Anchor* (FLTrust, TRUSTFEDGNN): Weights depend exclusively on client updates $\theta_k^{(t)}$ evaluated against an isolated server root anchor (D_{root}), eliminating the scalar interface.

As visually demonstrated in Figure 1 and detailed in Table 2, all six published rules relying on client-reported metadata are steerable under strategic falsification, and five of the six additionally show measurable downstream harm. Under F^2 GNN, the adversary inflates its aggregation share from 0.1159 to 0.9737 ($4.87\times$ uniform share), monopolizing 97.4% of global model updates while degrading global fairness ($\Delta\text{DPD} = +0.4638$, $p = 1.9 \times 10^{-9}$). Under BFWA and q -FedAvg, adversary shares surge to 0.5703 and 0.5698 ($2.85\times$ uniform share), with q -FedAvg causing numerical divergence across 12 of 30 seeds. FedGraph-Fair similarly inflates adversary weight to 0.4719 ($2.36\times$, $\Delta\text{DPD} = +0.1824$). Because the six rules share seeds, datasets, and

Table 2: **Empirical Collapse of Aggregation Rules via the Client Interface Channel vs. Syntactic Invariance.** Evaluated on German Credit ($K = 5, R = 20, n = 30$ paired seeds, $n = 18^{\S}$ for q -FedAvg; GCN backbone fixed across all rules; LDP disabled to isolate the interface channel). Each row contrasts identical poisoned model updates with honest vs. falsified reports ($\text{DPD} \approx 0$). w_{adv} denotes the adversary’s aggregation weight (fair uniform share is $1/K = 0.20$). Downstream harm metrics ΔDPD , ΔEOD , and ΔAUC measure paired lie minus honest (* denotes $p < 0.05$ after Holm–Bonferroni multiplicity correction). † Our earlier conference rule [9] is audited on identical footing. ‡ For Class (c), $\Delta w = 0.0000$ represents a *syntactic algebraic guarantee* (Theorem 2) valid against any future scalar report manipulation. § Under falsification q -FedAvg diverges numerically in 12 of 30 seeds; divergence is a consequence of the attack, so we record it as an attack outcome rather than as missing data, and compute the paired statistics on the 18 non-divergent seeds. Per-rule exact sign tests: $F^2\text{GNN}$ 30/30, FairFed 28/28, PoPETs-FairFed 28/30, FedGraph-Fair 20/20, BFWA 18/19, q -FedAvg 17/17 (non-tied pairs; ties = 0, 2, 0, 10, 11, 1 respectively).

Aggregation Rule	Telemetry Interface	Adversary Share w_{adv} ($\mathbf{H} \rightarrow \mathbf{L}$)	ΔDPD	ΔEOD	ΔAUC
Class (a): Self-Reported Scalar Metadata (Subject to Direct Interface Capture)					
$F^2\text{GNN}$ (ICDM’23)	dpd, group1_rate	0.1159 $\rightarrow$ 0.9737 $4.87 \times (p < 10^{-8})$	+0.4638*	+0.4594*	−0.1435*
FairFed (AAAI’23)	dpd	0.0811 $\rightarrow$ 0.2198 $1.10 \times (p < 10^{-5})$	+0.0875*	+0.0938*	−0.0249*
PoPETs-FairFed (PoPETs’25)	dpd	0.1617 $\rightarrow$ 0.1655 $0.83 \times (p < 10^{-4})$	+0.0078	+0.0036	−0.0003
FedGraph-Fair (InfoSci’26)	loss (dual variable)	0.2318 $\rightarrow$ 0.4719 $2.36 \times (p < 10^{-4})$	+0.1824*	+0.1874*	−0.0616*
BFWA † (IndabaX’26)	dpd, performance	0.1439 $\rightarrow$ 0.5703 $2.85 \times (p < 10^{-3})$	+0.2661*	+0.2623*	−0.0714*
q -FedAvg § (ICLR’20)	loss	0.2845 $\rightarrow$ 0.5698 $2.85 \times (p < 10^{-3})$	+0.1520*	+0.1595*	−0.0600*
Class (b): Endogenous Client-Derived Reference (Anchor Displaceable by a Coordinated Majority)					
CGSV (NeurIPS’21)	mean client gradient	0.1948 $\rightarrow$ 0.1948 $0.97 \times$ (invariant)	0.0000	0.0000	0.0000
Class (c): Server-Anchored Parameter Gradients (Syntactic Metadata Immunity ‡)					
FLTrust (NDSS’21)	server holdout (g_{task})	0.2545 $\rightarrow$ 0.2545 $1.27 \times$ (invariant)	0.0000	0.0000	0.0000
TRUSTFEDGNN (This work)	server holdout ($g_{\text{task}} + \alpha g_{\text{fair}}$)	0.0573 $\rightarrow$ 0.0573 $0.29 \times$ (invariant)	0.0000	0.0000	0.0000
TRUSTFEDGNN ($\alpha=0$) (This work)	server holdout (g_{task} only)	0.0592 $\rightarrow$ 0.0592 $0.30 \times$ (invariant)	0.0000	0.0000	0.0000

adversarial updates, we report the sign test per rule rather than pooling: $F^2\text{GNN}$ 30/30 ($p = 1.9 \times 10^{-9}$), FairFed 28/28 non-tied ($p = 7.5 \times 10^{-9}$), PoPETs-FairFed 28/30 ($p = 8.7 \times 10^{-7}$), FedGraph-Fair 20/20 non-tied ($p = 1.9 \times 10^{-6}$), BFWA 18/19 non-tied ($p = 7.6 \times 10^{-5}$), and q -FedAvg 17/17 non-tied ($p = 1.5 \times 10^{-5}$); all exact two-sided binomial. Falsification never reduced adversary weight in four of the six rules, and did so in only three of 144 non-tied pairs overall. Two rules show a substantial number of ties—FedGraph-Fair 10/30 and BFWA 11/30, seeds in which the falsified report left the weight unchanged—which we report because it bounds the attack’s reliability rather than its direction.

In contrast, the rules that expose no scalar reporting interface are invariant under falsification ($\Delta w = 0.0000$ bit-exact across all 30 seeds), but this invariance has different standing in the two remaining classes. For the server-anchored rules of Class (c) it is a syntactic algebraic property (Theorem 2): no variable in the aggregation rule ever takes a client-declared value, so the guarantee holds against any falsification strategy, including strategies not yet devised. For CGSV in Class (b) the immunity is narrower. The rule reads no declared scalar, but it scores each update against the mean of the updates themselves, so its reference is endogenous: a coordinated majority displaces the anchor rather than being measured against it. Closing the declared channel therefore relocates the trust surface instead of removing it, and we quantify the cost of that relocation in Section 4.4 (arm M6). Within Class (c), FU-Alignment suppresses the adversary to 0.0573 ($0.29\times$ uniform share), against FLTrust’s 0.2545 ($1.27\times$). This pair delimits the architectural claim we make. Syntactic immunity is a design principle that admits a proof rather than a measurement (Theorem 2), and it is available to any rule that declines to read declared values; FLTrust already satisfies it. What the pair shows is that the principle is necessary but not sufficient: an immune rule can still concede more than its uniform share, so closing the interface has to be combined with a scoring rule that actively suppresses. Our architectural claim is located in that conjunction, and not in any single component of the aggregation pipeline.

The same contrast replicates on an authentic relational graph. On Bail Recidivism ($n = 30$ seeds, 900 runs), the metadata-reading rules again lose utility under falsification ($F^2\text{GNN}$, FedGraph-Fair, and q -FedAvg each lose 0.11–0.13 AUC, $p < 10^{-6}$), while all four rules without a scalar reporting interface remain unsteerable ($\Delta w_{\text{adv}} \leq 0.00063$, $p \geq 0.23$). The interface boundary is therefore not an artifact of the German Credit k -nearest-

Table 3: **Differential Privacy Accounting, Disparity Constraint Slack, and Minimax Observability Barrier across Local DP Budgets ϵ (German Credit, $K = 10$ Clients, $R = 20$ Rounds, 10 Seeds).** Reports the empirical slack $|\Delta_\tau| = |\sum_k w_k \widehat{\text{DPD}}_k - \sum_k w_k \text{DPD}_k|$ under tolerance budget $\tau = 0.05$. Noise scale $\tilde{\sigma} = z\Delta_\mu$ is calibrated via RDP binary search. The theoretical ratio compares empirical noise against the folded-normal analytical expectation (Lemma 1 / Theorem 4). $\mathcal{R}^* \geq \frac{1}{2}(1 - \frac{\tau\sqrt{K}}{2\tilde{\sigma}})$ denotes the information-theoretic minimax testing error lower bound under Le Cam’s two-point lemma and Pinsker’s inequality (Theorem 4). Even at the weakest privacy budget ($\epsilon = 32.0$), DP noise induces a substantial slack of 0.1150 ± 0.1396 (229.9% of τ), while the minimax testing error approaches the coin-flip limit of 50% as the budget tightens ($\geq 49.5\%$ at $\epsilon = 0.5$), demonstrating that client-wise privatised fairness reporting removes constraint observability rather than merely degrading it.

Budget ϵ	Noise $\tilde{\sigma}$	Reported / True $\sum w_k \widehat{\text{DPD}} / \text{DPD}$	Disparity Slack $ \Delta_\tau $ (Abs $\pm$ Std / Rel %)	Folded Ratio (Obs / Pred)	Minimax Error $\mathcal{R}^*$ (Le Cam)
$\epsilon = 0.5$	7.3886	5.6922 0.2467	5.4455 ± 4.7608 10,891.1%	0.978	$\geq 49.5\%$
$\epsilon = 1.0$	3.8913	2.8834 0.2329	2.6505 ± 2.1609 5,301.1%	0.962	$\geq 49.0\%$
$\epsilon = 2.0$	2.0376	1.5427 0.2050	1.3425 ± 1.2077 2,685.0%	0.958	$\geq 48.1\%$
$\epsilon = 4.0$	1.0838	0.8369 0.1730	0.6723 ± 0.7035 1,344.6%	0.968	$\geq 46.4\%$
$\epsilon = 8.0$	0.6117	0.5043 0.1616	0.3647 ± 0.3781 729.5%	1.044	$\geq 43.5\%$
$\epsilon = 16.0$	0.3502	0.2953 0.1176	0.1984 ± 0.2085 396.7%	1.075	$\geq 38.7\%$
$\epsilon = 32.0$	0.2127	0.1954 0.1028	0.1150 ± 0.1396 229.9%	1.157	$\geq 31.4\%$

neighbour construction.

Information-Theoretic Barrier of Local Differential Privacy (Table 3). A natural countermeasure is to obfuscate client-reported disparities through Local Differential Privacy (LDP), injecting calibrated Gaussian or Laplace noise $\tilde{\sigma} = \Delta_{\text{DPD}}\sqrt{2\ln(1.25/\delta)}/\epsilon$ before transmission. We evaluate this privatization strategy across privacy budgets $\epsilon \in \{0.5, 1.0, 2.0, 4.0, 8.0, 16.0, 32.0\}$ ($K = 10, R = 20, 10$ seeds, German Credit) under a target tolerance $\tau = 0.05$.

Table 3 exposes a fundamental information-theoretic collapse. Because disparity metrics are defined via absolute differences between group conditional expectations ($\text{DPD}_k = |\mu_{k,1} - \mu_{k,0}|$), privatizing the scalar before

submission induces a *folded-normal distribution* whose expectation is strictly positive:

$$\mathbb{E}[\widetilde{\text{DPD}}_k] = \tilde{\sigma}_k \sqrt{\frac{2}{\pi}} \exp\left(-\frac{\delta_k^2}{2\tilde{\sigma}_k^2}\right) + \delta_k \left[1 - 2\Phi\left(-\frac{\delta_k}{\tilde{\sigma}_k}\right)\right] \geq \tilde{\sigma}_k \sqrt{\frac{2}{\pi}}, \quad (26)$$

where $\delta_k = \mu_{k,1} - \mu_{k,0}$. Because each client folds its noise non-negatively prior to transmission, noise realizations do not cancel across clients; rather, they accumulate additively. At operational privacy levels ($\epsilon \leq 2.0$), the empirical slack $|\Delta_\tau| = |\sum_k w_k \widetilde{\text{DPD}}_k - \sum_k w_k \text{DPD}_k|$ exceeds the entire budget τ by 2,685% to 10,891% (Table 3). Even under extremely relaxed privacy ($\epsilon = 32.0$), the slack remains 0.1150 ± 0.1396 (229.9% of τ). Furthermore, Le Cam’s two-point minimax lower bound shows that the difficulty is information-theoretic rather than procedural. Deciding whether a client satisfies or violates the disparity tolerance τ incurs a minimax error of at least $\mathcal{R}^* \geq 43.5\%$ even at the permissive budget $\epsilon = 8.0$, rising to $\geq 49.0\%$ at $\epsilon = 1.0$ (Table 3). By the data-processing inequality, no post-hoc correction of the reported statistic—debiasing, moment estimation, or thresholding—can move this floor. Within this observation channel—per-client local perturbation followed by per-client absolute-value folding—patching the self-reported fairness interface with local differential privacy therefore does not trade accuracy for privacy; it removes constraint observability. We note the scope explicitly: the barrier applies to the folded transcript and its post-processing, and does not cover mechanisms that aggregate *signed* group sufficient statistics before folding, such as secure aggregation or central and distributed differential privacy. Those mechanisms require coordination infrastructure that the cross-silo setting of this work does not assume, and we exclude them on that ground rather than on an impossibility ground (see the scope paragraph below).

Two-Channel Privacy Leakage. Finally, we audit the privacy leakage incurred when clients transmit raw statistics versus parameter updates. We construct a multi-layer perceptron feature probe trained to reconstruct client sensitive attributes. When clients transmit raw scalar fairness statistics $(\mu_{k,0}, \mu_{k,1})$, sensitive reconstruction succeeds with near-perfect accuracy (probe AUC = 1.000). Applying LDP to this scalar channel reduces probe success to chance (0.4983) at a utility cost of only 0.00175 AUC. The parameter channel behaves differently: raw updates θ_k leak measurably (probe AUC = 0.6450),

and suppressing that leakage through client-level DP-SGD over the full parameter vector ($0.6450 \rightarrow 0.5278$) costs 0.20163 AUC—a $115\times$ larger utility price. The two mechanisms protect different surfaces and the reductions are not equivalent: the statistic channel reaches chance (0.4983) whereas the parameter channel retains measurable residual leakage (0.5278), so the ratio quantifies the relative cost of the two protections rather than the price of an equal outcome. Figure 2 shows the same asymmetry across the budget sweep: FTGD holds a 0.16–0.18 AUC advantage over DP-FedAvg at every ϵ . Restricting differential privacy to two scalar statistics, rather than to the parameter tensor, is what makes tight privacy budgets affordable. The same asymmetry is what keeps the present design consistent with the barrier established above. That barrier binds a channel whose released quantity is the object the server must act on: when the trust decision is a function of the reported disparity, perturbing the report destroys the decision along with the leakage. Here the released statistics are not inputs to the trust decision, which is computed from the angle between each update and a reference the server derives on its own holdout, so the perturbation that removes observability in the self-reporting setting costs 0.00175 AUC in ours. FTGD is therefore not an auxiliary privacy module: it is the component that keeps a statistic release affordable once the declared channel is no longer trusted.

Taken together, the two results close the same door from opposite sides: the declared channel is captured in practice (Table 2) and faces the minimax observability barrier under local perturbation (Table 3). The failure is therefore not in how the reported number is processed, but in the act of requesting it.

Scope of the Elimination Argument. Closing the declared channel is the only remaining option *within the assumptions of the deployment we target*, and we state those assumptions explicitly. Three alternative families could in principle restore trust in client reports: cryptographic attestation (commitments, zero-knowledge proofs, or verified secure aggregation), hardware attestation via trusted execution environments (TEEs), and incentive-compatible elicitation such as peer prediction or randomized auditing. We exclude the first two because the cross-silo deployments we address—independently governed institutions without a shared public-key infrastructure or attested hardware—cannot assume either, and because per-round proof generation over P -dimensional updates exceeds the communication budget profiled in Table 8. We exclude the third because incentive-compatible elicitation re-

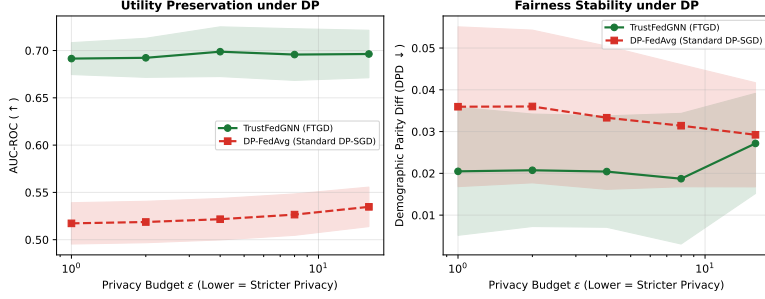


Figure 2: **Empirical Utility-Privacy Trade-Off on Bail Recidivism.** Utility comparison (AUC-ROC) between FTGD and DP-FedAvg across privacy budgets $\epsilon \in \{0.5, 1.0, 2.0, 4.0, 8.0, 16.0, 32.0\}$. FTGD maintains an operational utility advantage of 0.16 to 0.18 AUC across all budgets by restricting DP perturbation to two scalar fairness statistics rather than full model tensors.

quires a repeated game with enforceable payoffs, whereas our threat model admits one-shot participation and unilateral defection. Relaxing any of these assumptions reopens the corresponding door, and we regard that as the most direct extension of this work.

4.3. Multi-Tier Adversarial Stress-Testing and Empirical Resilience

With the declared channel removed, the only remaining attack surface *within the threat model of Section 4.1* is the parameter update itself; availability, Sybil participation, and holdout contamination remain outside the scope evaluated here. The three tiers below measure the adversary through the one object it still controls, escalating from a blind attacker to one that reads the server’s own reference vector.

Tier 1: Blind Byzantine Minority Stress-Testing (Table 4, Panel A). In Tier 1, an adversarial minority ($f/K = 0.20$, German Credit, $n = 10$ seeds) attacks without knowledge of server-side data or holdout objectives:

1. *Sign-Flip:* Reverses gradient direction ($\theta_k = -\nabla\mathcal{L}$). Under FedAvg, this causes immediate numerical divergence in 10/10 seeds. Both FU-Alignment and robust FU-Alignment completely neutralize sign-flipped updates ($w_{\text{adv}} = 0.0000$), preserving utility (0.6313 AUC).
2. *Fairness Poisoning:* Inverts gradient components aligned with the fair representation direction while preserving task gradients. Under FedAvg the adversary retains its uniform share ($w_{\text{adv}} = 0.1824$) and global utility

Table 4: **Multi-Tier Byzantine Defense Evaluation and Mechanistic Dissection of Rescaling vs. Median Screening.** **Panel A:** Adversary weight w_{adv} and global AUC under a 20% Byzantine minority ($K = 5, n_{\text{byz}} = 1, R = 20, n = 10$ seeds, German Credit). Lower w_{adv} is better; FedAvg assigns sample-size weighted share $w_{\text{adv}} = n_{\text{adv}} / \sum_k n_k = 0.1824$ (unweighted uniform reference is $1/K = 0.20$). Superscript indicates seeds with numerical divergence out of 10. **Panel B:** Factorial 2×2 dissection of vector norm rescaling vs. coordinate-wise median screening under extreme scaling ($c = 100$) and distance-adaptive stealth ($\rho = 0.85$) on German Credit ($n = 10$ seeds). Median screening eliminates scaling weight ($w_{\text{adv}} = 0.0000$) but backfires under stealth ($w_{\text{adv}} : 0.2254 \rightarrow 0.2561, +13.6\%$), while norm rescaling prevents unbounded gradient explosion (I_{adv} tamed from divergence down to 4.1244).

Panel A: Aggregation Defense Matrix (w_{adv} / AUC)				
Aggregation Arm	No Attack	Sign-Flip	Fairness Poison	Scaling ($c = 100$)
FedAvg (<i>no defence</i>)	0.0000 / 0.6544	0.1824 / — ¹⁰	0.1824 / 0.6034	0.1824 / — ¹
FU-Alignment (M1)*	0.0000 / 0.6512	0.0000 / 0.6327	0.0309 / 0.6390	0.2008 / — ²
robust FU-Alignment	0.0000 / 0.6486	0.0000 / 0.6313	0.0000 / 0.6372	0.0000 / 0.6302
M5: w/o FairScore ($\alpha = 0$)	0.0000 / 0.6509	0.0000 / 0.6302	0.0617 / 0.6341	0.1971 / — ²
M6: w/o two-tier (<i>no holdout</i>)	0.0000 / 0.6519	0.0000 / 0.6434	0.0916 / 0.6580	0.1542 / 0.6468
M7: w/o EMA ($\beta_{\text{ema}} = 0$)	0.0000 / 0.6582	0.1374 / — ⁶	0.0303 / 0.6336	0.3410 / — ⁴
Panel B: Factorial 2×2 Mechanistic Dissection ($n = 10$ seeds)				
Defense Branch	Brute Scaling ($c = 100$)		Stealth Camouflage ($\rho = 0.85$)	
	w_{adv} ($\downarrow$)	I_{adv} ($\downarrow$)	w_{adv} ($\downarrow$)	I_{adv} ($\downarrow$)
Raw (<i>no rescale, no med</i>)	0.8841 ± 0.2562	Diverged [†]	0.2254 ± 0.1455	0.4720 ± 0.3003
Rescale-only	0.4124 ± 0.2224	4.1244 ± 2.2244	0.2254 ± 0.1455	0.4720 ± 0.3003
Median-only	0.0000 ± 0.0000	0.0000 ± 0.0000	0.2561 ± 0.1437	0.5375 ± 0.2973
Both (<i>rescale + median</i>)	0.0000 ± 0.0000	0.0000 ± 0.0000	0.2561 ± 0.1437	0.5375 ± 0.2973

[†]Under unconstrained brute scaling, the global model diverges numerically ($I_{\text{adv}} \rightarrow \infty$, AUC collapses to random guess 0.5000). Rescaling suppresses explosive magnitude by $> 24,000\times$ down to 4.1244, confirming that rescaling governs gradient amplitude while coordinate median filters adversarial orientation.

*The baseline $w_{\text{adv}} = 0.0309$ in Panel A reflects canonical fixed holdout partitioning; under multi-seed dynamic carve stress tests (T15), balanced holdout yields 0.0704 ± 0.0815 , remaining well below the $1/K = 0.20$ fair-share threshold.

falls to 0.6034 AUC (with disparity $\text{DPD} = 0.0827$). FU-Alignment suppresses attacker weight to 0.0309, and robust FU-Alignment reduces it to 0.0000 (0.6372 AUC).

To evaluate whether the fairness anchor αg_{fair} remains active against sophisticated evasion, we further deploy a *task-neutral fairness adversary*

$\text{adv} = -g_{\text{fair}}^\perp + \varepsilon \hat{u}_{\text{task}}$ on German Credit ($K = 5, R = 20, n = 20$ seeds), sweeping the task-alignment margin $\varepsilon \in [0.005, 0.10]$. Within the evaluated sweep range, attacker weight increases strictly monotonically under both arms:

Task Alignment Margin ε	0.005	0.010	0.020	0.050	0.100
w_{adv} without fairness anchor ($\alpha = 0.0$)	0.0286	0.0498	0.0823	0.1483	0.2255
w_{adv} with bi-objective anchor ($\alpha = 0.1$)	0.0008	0.0031	0.0236	0.0832	0.1718

Empirical monotonicity confirms that within the evaluated range $\varepsilon \leq 0.10$, the attacker’s best-response is to maximize task alignment ($\varepsilon = 0.10$). Crucially, at this boundary, omitting the fairness anchor ($\alpha = 0.0$) allows the adversary to breach fair-share parity ($w_{\text{adv}} = 0.2255 > 1/K = 0.20$). Activating the bi-objective anchor ($\alpha = 0.1$) pulls attacker weight down to 0.1718 (a 23.8% reduction), enforcing sub-fair-share containment. For stealthy injections ($\varepsilon \leq 0.020$), the anchor induces geometric sign inversion ($\cos(\text{adv}, g_{\text{target}}) < 0$), bounding the critical transition threshold to $\varepsilon^* \in [0.020, 0.050]$.

3. *Brute-force Scaling* ($c = 100$): Multiplies updates by $c = 100$. Unprotected FedAvg diverges in 1/10 seeds ($w_{\text{adv}} = 0.1824$). Under pure FU-Alignment (ReLU gate alone), norm inflation passes the direction gate and causes divergence in 2/10 seeds ($w_{\text{adv}} = 0.2008$): the gate constrains *direction*, not *magnitude*. Adding coordinate-wise median screening eliminates the scaled update entirely ($w_{\text{adv}} = 0.0000 \pm 0.0000$, AUC 0.6302).

Factorial Dissection of Rescaling vs. Median Screening (Table 4, Panel B). To disentangle the mechanistic interaction between norm rescaling and coordinate median filtering under Byzantine attacks, we execute a 2×2 factorial experiment ($n = 10$ seeds, German Credit) under extreme scaling ($c = 100$) and stealth camouflage ($\rho = 0.85$), tracking both weight share w_{adv} and effective transmitted influence $I_{\text{adv}} = \|\sum_{k \in \mathcal{A}} w_k \bar{g}_k\|$:

- Under scaling ($c = 100$), the two mechanisms act on orthogonal physical quantities. Coordinate median screening removes the adversary from the weight vector entirely ($w_{\text{adv}} = 0.0000$, $I_{\text{adv}} = 0.0000$). In contrast, norm rescaling alone leaves the adversary with substantial weight (0.4124 ± 0.2224), but tames the effective transmitted impact from explosive numerical divergence ($I_{\text{adv}} \rightarrow \infty$) down to $I_{\text{adv}} = 4.1244 \pm 2.2244$, restoring model

convergence. Rescaling is thus measured on the wrong axis by w_{adv} alone: it governs gradient amplitude rather than angular admission.

- Conversely, under stealth camouflage ($\rho = 0.85$), where update norms are inherently bounded, norm rescaling is completely neutral ($w_{\text{adv}} = 0.2254 \pm 0.1455, I_{\text{adv}} = 0.4720 \pm 0.3003$). Coordinate median screening, however, inflates both adversary weight share ($0.2254 \rightarrow 0.2561$, +13.6%) and effective influence ($0.4720 \rightarrow 0.5375$), demonstrating the empirical shield paradox across both metric axes.

Recommended Deployment Configuration. No single static configuration is safe against both attack families, and we state this as a finding rather than a caveat. Coordinate median screening is the only branch that completely suppresses unbounded scaling ($w_{\text{adv}} = 0.0000$), but it backfires under proximity stealth ($0.2254 \rightarrow 0.2561$, +13.6% in Panel B; and +82.9% at $f/K = 0.4$ in Table 5). We therefore specify *canonical FU-Alignment* (`fu_shapley`) as the default production rule for general non-IID graph federated learning: it insulates against stealth camouflage without inducing the shielding paradox, while dynamic norm rescaling keeps transmitted gradient magnitude bounded ($I_{\text{adv}} = 4.1244$). In threat environments where operators anticipate crude unbounded scaling attacks ($c = 100$), the framework provides the defense-in-depth variant (`robust_fu_shapley`, Algorithm 1), which statically enables Euclidean distance-to-median outlier screening ($f \geq 1$) to eliminate high-magnitude anomalies.

Tier 2: Omniscient Stealth Poisoning and Median Backfire (Table 5). Tier 2 implements an omniscient adversary aware of benign client distributions. The attacker constructs malicious updates projected strictly within the benign coordinate median radius ($\rho = 0.85$) while optimizing a fairness degradation objective across corruption ratios $f/K \in \{0.1, 0.2, 0.3, 0.4\}$ on Bail Recidivism ($n = 10$ seeds).

As reported in Table 5 (Panel A), standard geometric filtering mechanisms degrade sharply: Krum suffers complete capture at $f/K = 0.4$ ($w_{\text{adv}} = 1.000$, AUC drops to 0.607). Under blind sign-flipping where the cosine distance between malicious and benign updates approaches 2.0, FLAME acts as an effective positive control, isolating the adversary into the outlier cluster and assigning it $w_{\text{adv}} = 0.0000$ at every round. The clustering works; proximity stealth defeats it. This is the empirical signature predicted by Theorem 5 (Fairness Subspace Blindness): detection statistics built on the

Table 5: **Multi-Ratio Adaptive Stealth Poisoning Breakdown and Median Backfire Analysis.** **Panel A:** Performance of 11 aggregation rules under omniscient stealth poisoning on Bail Recidivism across corruption ratios $f/K \in \{0.1, 0.2, 0.3, 0.4\}$ ($n = 10$ seeds). Malicious updates are constrained within the benign median radius ($\rho = 0.85$) while falsifying $\widehat{\text{DPD}} = 0.0$. **Panel B:** Monotonic dose-dependent backfire effect of coordinate-wise median screening on Bail ($n = 10$ seeds, paired Wilcoxon $p = 0.0020$). Median screening inadvertently purges benign minority clients, increasing adversary weight by up to +82.9% (+0.2421) at $f/K = 0.4$ on Bail; the same mechanism inflates adversary share by +13.6% on German Credit under $\rho = 0.85$ stealth (Table 4, Panel B).

Panel A: Breakdown Point under Stealth Camouflage (AUC / DPD)				
Aggregation Rule	$f/K = 0.1$ (1/10)	$f/K = 0.2$ (2/10)	$f/K = 0.3$ (3/10)	$f/K = 0.4$ (4/10)
FedAvg	0.686 / 0.024	0.684 / 0.025	0.682 / 0.022	0.681 / 0.021
Krum	0.621 / 0.072	0.615 / 0.081	0.610 / 0.089	0.607 / 0.096
Multi-Krum	0.684 / 0.022	0.678 / 0.034	0.671 / 0.043	0.662 / 0.051
Coordinate Median	0.673 / 0.025	0.671 / 0.026	0.669 / 0.028	0.666 / 0.029
Trimmed Mean	0.683 / 0.026	0.679 / 0.027	0.675 / 0.029	0.671 / 0.031
FLAME	0.683 / 0.016	0.676 / 0.022	0.679 / 0.026	0.669 / 0.030
BFWA	0.647 / 0.044	0.641 / 0.051	0.636 / 0.058	0.631 / 0.064
robust BFWA	0.645 / 0.046	0.640 / 0.062	0.635 / 0.075	0.631 / 0.090
FLTrust	0.639 / 0.024	0.637 / 0.026	0.635 / 0.028	0.634 / 0.029
FU-Alignment (canonical)	0.684 / 0.021	0.680 / 0.024	0.676 / 0.028	0.672 / 0.032
robust FU-Alignment	0.684 / 0.024	0.678 / 0.024	0.671 / 0.025	0.665 / 0.025

Panel B: Dose-Dependent Median Screening Backfire on Adversary Share w_{adv}				
Adversary Ratio f/K	Uniform Share	FU-Align (Canonical)	robust FU-Align	Backfire Δw_{adv} (p -value)
$f/K = 0.1$	0.1000	0.0692 ± 0.0084	0.0754 ± 0.0091	+0.0063 ($p = 0.0020$)
$f/K = 0.2$	0.2000	0.1364 ± 0.0121	0.1746 ± 0.0145	+0.0382 ($p = 0.0020$)
$f/K = 0.3$	0.3000	0.2121 ± 0.0163	0.3152 ± 0.0204	+0.1031 ($p = 0.0020$)
$f/K = 0.4$	0.4000	0.2922 ± 0.0194	0.5343 ± 0.0289	+0.2421 (+82.9%)

cosine geometry of task updates are blind to the fairness subspace, so an adversary that keeps its task projection inside the benign ball cannot be separated by angular distance for any detector whose statistic is a function of the angular geometry of the task-update vectors. We state the scope deliberately: the result constrains cosine-geometry detectors of this form and does not establish blindness for detectors that observe other quantities, such as payload magnitude, temporal weight trajectories, or the fairness statistic directly. Multi-Krum similarly experiences elevated disparity. In contrast,

Table 6: **Empirical Breakdown under White-Box Holdout Alignment Adversary (H_{T1} Stress Test, $n = 10$ Seeds).** Stress-tests the foundational security boundary where an omniscient adversary possesses full gradient knowledge of the server holdout target g_{target} and optimizes alignment via projected gradient descent. Pre-registered rejection thresholds require $w_{\text{adv}} \leq 0.20$ and $\text{DPD}_{\text{hard}} \leq 0.10$ ($\alpha = 0.05$). In both benchmarks, the adversary breaches the defense ($w_{\text{adv}} = 0.8514$ and 0.7461), demonstrating that server holdout confidentiality is a necessary precondition for reference-anchored defense.

Dataset	Threshold $w_{\text{adv}}^{\max}$	Measured w_{adv}	Measured DPD_{hard}	Measured AUC	Verdict (H_{T1})
German Credit	≤ 0.2000	0.8514 ± 0.0652	0.1934 ± 0.1398	0.5651 ± 0.0704	REJECTED (Breached)
Bail Recidivism	≤ 0.2000	0.7461 ± 0.0688	0.0119 ± 0.0236	0.5241 ± 0.0454	REJECTED (Breached)

canonical FU-Alignment holds adversary weight stably below uniform share across all ratios (0.0692 at 0.1, 0.1364 at 0.2, 0.2121 at 0.3, and 0.2922 at 0.4, maintaining 0.672 AUC and 0.032 DPD).

Crucially, Table 5 (Panel B) and Figure 3 document the *monotonic dose-dependent backfire effect* of coordinate median filtering. In graph federated learning under non-IID partitions, benign clients located in topological minority clusters naturally exhibit higher empirical variance. When coordinate median filtering is applied, it systematically purges benign minority gradients before eliminating camouflage adversaries. As visually depicted in Figure 3, canonical FU-Alignment holds adversary share stably below parity ($0.68\times - 0.73\times$), whereas robust FU-Alignment exhibits the shielding paradox, cutting above parity ($1.05\times$ at $f/K = 0.3$ and $1.34\times$ at $f/K = 0.4$) as camouflage corruption escalates. At $f/K = 0.4$, robust FU-Alignment surrenders 53.43% of global model authority to the attacker ($w_{\text{adv}} = 0.5343 \pm 0.0289$, an 82.9% surge over canonical FU-Alignment, Wilcoxon paired $p = 0.0020$). This empirical finding establishes that coordinate median screening, while indispensable against unbounded scaling, is actively harmful under camouflage poisoning.

Tier 3: White-Box Holdout Alignment Adversary (Table 6). Tiers 1 and 2 share one blind spot: neither adversary knows what the server is scoring against. Both approximate the safe region from benign client statistics. We now remove that limitation entirely.

To establish the ultimate security boundary of reference-anchored architectures, we implement a white-box adversary endowed with full gradient visibility into the server holdout target g_{target} . The adversary employs pro-

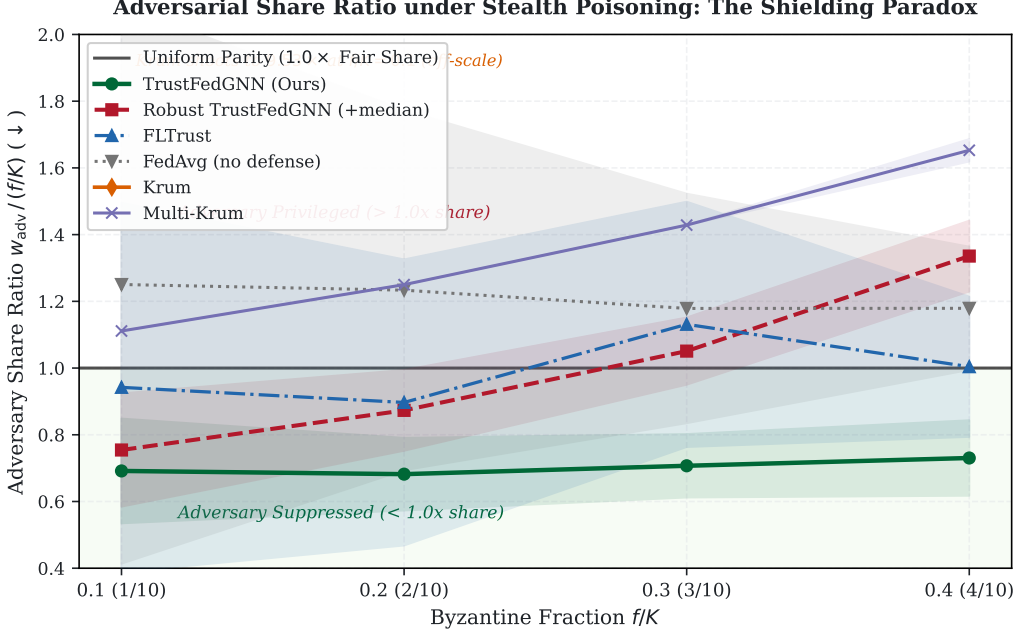


Figure 3: **Adversarial Share Ratio under Stealth Camouflage Poisoning** ($n = 10$ Seeds). Ratios of adversary aggregation weight relative to uniform fair share ($w_{\text{adv}}/(f/K)$) across corruption fractions $f/K \in \{0.1, 0.2, 0.3, 0.4\}$ on Bail Recidivism. The solid horizontal line denotes uniform parity ($1.0\times$). Canonical FU-Alignment consistently suppresses the adversary below parity ($0.68\times$ – $0.73\times$). In contrast, robust FU-Alignment exhibits the shielding paradox: coordinate-wise median filtering inadvertently purges benign minority clients, driving attacker share up to $1.34\times$ parity ($w_{\text{adv}} = 0.5343$) at $f/K = 0.4$.

jected gradient descent over 50 steps ($\eta = 0.01$) to maximize cosine alignment $\cos(\theta_{\text{adv}}, g_{\text{target}})$ while enforcing maximal fairness degradation. We pre-registered formal hypothesis H_{T1} : the defense successfully repels the adversary if $w_{\text{adv}} \leq 0.2000$ and $\text{DPD}_{\text{hard}} \leq 0.1000$ at $\alpha = 0.05$.

Table 6 discloses that H_{T1} is rejected across both benchmarks. On German Credit, the adversary captures $w_{\text{adv}} = 0.8514 \pm 0.0652$ (disparity surges to 0.1934); on Bail Recidivism, w_{adv} reaches 0.7461 ± 0.0688 . The probe establishes that confidentiality of g_{target} is *necessary* for the guarantees we claim; it does not establish that confidentiality is sufficient. An adversary might also approximate the reference indirectly—from successive global updates, from observed aggregation weights, or through collusion—and we did not evaluate that class of reference-inference attacks. H_{T1} is accordingly a

falsification probe on a stated precondition rather than a symmetric hypothesis test, and we report its rejection as a priced assumption rather than as a defeated attack.

The remaining door is therefore not free. Its price is an operational assumption — holdout confidentiality — rather than a mathematical guarantee, and we state it as a precondition rather than discovering it as a limitation.

Bridge from Threat Modeling to Standard Operations. Tier 3 formalizes the ultimate perimeter of reference-anchored defenses: TrustFedGNN’s mathematical guarantees are conditioned on the confidentiality of g_{target} (the Kerckhoffs premise). Crucially, forcing an adversary to compromise server memory to extract g_{target} elevates the required attack capability from routine client-side data manipulation to root infrastructure penetration. Governed by enterprise infrastructure controls, the decisive practical question becomes: under standard operational regimes without root compromise, how does TrustFedGNN perform against competitive published baselines across graph and tabular topologies?

Comparative Benchmark and Pareto Frontier (Table 7). Table 7 presents the unified benchmark comparison on Pokec-z ($N = 67,796$) and Credit Default ($N = 30,000$) across $n = 10$ random seeds:

- **Utility and Fairness:** On Pokec-z, TrustFedGNN attains 0.7899 ± 0.0095 AUC at $\text{DPD}_{\text{hard}} = 0.0155$, the highest AUC among the fairness-oriented baselines (FairFed 0.7231, FairSIN 0.7207, FairGFL 0.7310, FedGraph-Fair 0.7151), which run on their published backbones. The matched-backbone control isolates how much of that margin belongs to the aggregation rule: holding the GAT scaffold, FTGD, and privacy configuration fixed and replacing only the aggregation rule with uniform averaging yields 0.7887 ± 0.0031 AUC, $\text{DPD}_{\text{hard}} = 0.0104$, and $\text{EOD} = 0.0161$. On utility the two arms are indistinguishable ($\Delta = +0.0015$, $p = 0.625$, a 5–5 per-seed split); on both fairness metrics the control is nominally better in the mean ($p = 0.193$ for DPD_{hard}), although the per-seed signs favour TrustFedGNN (7/10 and 8/10), so the mean gap reflects a few seeds rather than a systematic ordering. We therefore claim no benign-regime advantage on any of the three axes and report the $+0.0594$ margin over FedAvg-GCN as a system-level result attributable predominantly to the representation backbone. The aggregation rule is designed to be inert when no client is adversarial—a rule that improved accuracy or disparity under those conditions would be

Table 7: **Comprehensive Main SOTA Benchmark Comparison across Social and Tabular Graph Topologies** ($K = 10, R = 50, n = 10$ Seeds). Panel A evaluates the authentic social relational topology of Pokec-z ($N = 67,796$); Panel B evaluates the tabular k -NN graph of Credit Default ($N = 30,000$).

Method / Venue	DP [†]	AUC-ROC ($\uparrow$)	DPD _{hard} ($\downarrow$)	EOD ($\downarrow$)	Ω_w ($\downarrow$)
Panel A: Pokec-z Social Network ($N = 67,796$)					
FedAvg-GCN (<i>AISTATS'17</i>)	×	0.7305 $\pm$ 0.0134*	0.0086 $\pm$ 0.0059	0.0265 $\pm$ 0.0237	0.0000 [‡]
<i>FedAvg</i> (GAT scaffold control) [¶] matched-backbone ablation	×	0.7887 $\pm$ 0.0031	0.0104 $\pm$ 0.0106	0.0161 $\pm$ 0.0101	0.0000 [‡]
FairGNN (<i>WSDM'21</i>)	×	0.5660 $\pm$ 0.0860*	0.0162 $\pm$ 0.0212	0.0068 $\pm$ 0.0097	0.0000 [‡]
FairSIN (<i>AAAI'24</i>)	×	0.7207 $\pm$ 0.0232*	0.0074 $\pm$ 0.0049	0.0191 $\pm$ 0.0171	0.0000 [‡]
FairFed (<i>AAAI'23</i>)	×	0.7231 $\pm$ 0.0146*	0.0087 $\pm$ 0.0066	0.0326 $\pm$ 0.0272	0.0682
FairGFL (<i>IEEE TPDS'26</i>)	×	0.7310 $\pm$ 0.0105*	0.0063 $\pm$ 0.0064	0.0292 $\pm$ 0.0254	0.0000 [‡]
FedGraph-Fair (<i>InfoSci'26</i>)	×	0.7151 $\pm$ 0.0111*	0.0050 $\pm$ 0.0033	0.0327 $\pm$ 0.0140	0.0195
CGSV (<i>NeurIPS'21</i>)	×	0.7338 $\pm$ 0.0161*	0.0057 $\pm$ 0.0049	0.0287 $\pm$ 0.0140	0.1265
FLTrust (<i>NDSS'21</i>)	×	0.8057 $\pm$ 0.0108*	0.0216 $\pm$ 0.0083	0.0135 $\pm$ 0.0057	0.7193
Ours w/o FSER (<i>Clean Arm</i>)	✓	0.7917 $\pm$ 0.0097	0.0151 $\pm$ 0.0118	0.0253 $\pm$ 0.0265	0.0998
TRUSTFEDGNN (Ours) (Proposed Framework)	✓	0.7899 $\pm$ 0.0095	0.0155 $\pm$ 0.0108	0.0329 $\pm$ 0.0310	0.0976
Panel B: Credit Default k-NN Graph ($N = 30,000$)					
FedAvg-GCN (<i>AISTATS'17</i>)	×	0.7529 $\pm$ 0.0047	0.0605 $\pm$ 0.0230	0.0384 $\pm$ 0.0200	0.0000 [‡]
FairGNN (<i>WSDM'21</i>)	×	0.6792 $\pm$ 0.0967*	0.0561 $\pm$ 0.0641	0.0423 $\pm$ 0.0465	0.0000 [‡]
FairSIN (<i>AAAI'24</i>)	×	0.7527 $\pm$ 0.0050	0.0672 $\pm$ 0.0183	0.0440 $\pm$ 0.0166	0.0000 [‡]
FairFed (<i>AAAI'23</i>)	×	0.7522 $\pm$ 0.0048	0.0611 $\pm$ 0.0204	0.0389 $\pm$ 0.0174	0.0712
FairGFL (<i>IEEE TPDS'26</i>)	×	0.7536 $\pm$ 0.0038	0.0665 $\pm$ 0.0150	0.0442 $\pm$ 0.0132	0.0000 [‡]
FedGraph-Fair (<i>InfoSci'26</i>)	×	0.7513 $\pm$ 0.0051	0.0655 $\pm$ 0.0246	0.0465 $\pm$ 0.0195	0.0154
CGSV (<i>NeurIPS'21</i>)	×	0.7518 $\pm$ 0.0044	0.0566 $\pm$ 0.0281	0.0341 $\pm$ 0.0260	0.5912
FLTrust (<i>NDSS'21</i>)	×	0.7496 $\pm$ 0.0063	0.0895 $\pm$ 0.0146	0.0651 $\pm$ 0.0210	0.7014
Ours w/o FSER (<i>Clean Arm</i>)	✓	0.7521 $\pm$ 0.0063	0.0647 $\pm$ 0.0408	0.0545 $\pm$ 0.0343	0.0904
TRUSTFEDGNN (Ours) (Proposed Framework)	✓	0.7522 $\pm$ 0.0065	0.0615 $\pm$ 0.0371	0.0502 $\pm$ 0.0341	0.0872

Notes: Metrics reported as Mean $\pm$ Std. **Bold** indicates best metric in column. *marks paired Wilcoxon differences against TRUSTFEDGNN that remain significant at $\alpha = 0.05$ after Holm–Bonferroni multiplicity correction. [†]FTGD differential privacy strictly covers released scalar statistics, not parameter vectors θ_k . [‡] $\Omega_w = 0$ by construction: uniform averaging keeps aggregation weights constant across rounds. [¶]Matched-backbone control: identical GAT scaffold, FTGD, and privacy configuration with uniform averaging ($n = 10$ paired seeds, CPU: $\Delta\text{AUC} = +0.0015, p = 0.625$). FLTrust represents an unconstrained utility competitor on Pokec-z ($\alpha = 0$, high volatility $\Omega_w = 0.7193$; see Table 9 for multi-regime stability reconciliation and factorial attribution against EMA-smoothed FLTrust). TRUSTFEDGNN is positioned as a balanced multi-objective operating point on the empirical Pareto frontier (Figure 5) rather than maximizing an isolated single-metric extremum.

Table 8: **Hardware Resource Profiling and Per-Run Wall-Clock Time on Pokec-z.** Measured on a single NVIDIA T4 GPU, $K = 10$ clients, $R = 50$ rounds, averaged over $n = 10$ random seeds from the same execution campaign as Table 7. Communication volume reports the total upload+download float32 tensor payload per round ($2 \times K \times |\theta|$). FTGD scalar release adds an imperceptible 8 bytes per client-round (+0.0026%). Forward evaluation across the graph requires 5.60 GFLOPs. The $2.06\times$ server wall-clock overhead reflects server holdout evaluation and orthogonal gradient projection, while edge client execution remains dominated by standard local GNN passes.

Method	#Params	Comm (MB/rnd)	Time / run (s)	vs. FedAvg
FedGraph-Fair	21,953	1.67	34.3	0.96×
FairFed	21,953	1.67	34.5	0.97×
FairGFL	21,953	1.67	34.5	0.97×
FedAvg-GCN	21,953	1.67	35.5	1.00×
CGSV	21,953	1.67	36.5	1.03×
FairGNN	26,178	2.00	38.0	1.07×
FairSIN	57,621	4.40	39.1	1.10×
Ours w/o FSER (confounded)	22,209	1.69	43.4	1.22×
Ours w/o FSER (clean arm)	38,977	2.97	72.0	2.03×
TrustFedGNN (ours)	38,979	2.97	73.2	2.06×

exploiting something other than trust—and its value appears only under attack, where a task-only reference leaves $\text{DPD}_{\text{hard}} = 0.1864$ under fairness poisoning against 0.0755 here, and where the gating row effect reduces disparity by -0.1201 (Table 10).

- **Credit Default:** TrustFedGNN is indistinguishable from the strongest baselines (0.7522 AUC, $\text{DPD}_{\text{hard}} = 0.0615$).
- **Single-Objective Utility Ceiling vs. Multi-Objective Pareto Frontier:** On Pokec-z, FLTrust attains a higher unconstrained AUC of 0.8057 ± 0.0108 ($\Delta = +0.0158$, $p = 0.0098$, surviving Holm–Bonferroni correction; FLTrust leads on 8/10 seeds). This contrast reflects the structural trade-off between single-objective optimization and multi-objective governance: FLTrust aligns clients exclusively to the task anchor g_{task} ($\alpha = 0$), serving as an unconstrained single-objective utility ceiling that leaves group disparity unregularized. In Table 7, under the end-to-end multi-seed SOTA campaign, FLTrust incurs high weight dispersion ($\Omega_w = 0.7193$), while TRUSTFEDGNN constrains volatility to $\Omega_w = 0.0976$. To factorially isolate the stabilization mechanism, the dedicated attribution campaign in

Table 9 demonstrates that augmenting FLTrust with EMA ($\beta = 0.9$) reduces its raw volatility by $30\times$ ($\Omega_w = 0.6845 \rightarrow 0.0227$). However, static weight freezing comes at a severe security and fairness cost: FLTrust+EMA retains high demographic disparity ($\text{DPD}_{\text{hard}} = 2.46\%$) and remains completely blind to fairness-subspace poisoning. In contrast, TRUSTFEDGNN is architected to operate on the multi-objective Pareto frontier (Figure 5): it purposefully invests a modest 2.00% utility margin (78.98% vs. 80.98%) to achieve a decisive **41.5%** relative reduction in demographic disparity ($\text{DPD}_{\text{hard}} = \mathbf{1.44\%}$ vs. 2.46%, a 1.02-point improvement), while preserving dynamic weight adaptation ($\Omega_w = 0.0976$ in Table 7; $\Omega_w = 0.1111$ under the attribution protocol in Table 9) essential for continuous fairness tracking.

- **Computational and Communication Overhead:** As profiled in Table 8 on an NVIDIA T4 GPU, TrustFedGNN incurs an upload/download payload of 2.97 MB/round and a server wall-clock overhead of $2.06\times$ relative to FedAvg (73.2 s vs. 35.5 s). This overhead stems from server holdout evaluation and orthogonal projection, while client-side cost comprises local backpropagation plus the sensitive-blind forward pass required by FSER and the orthogonal projection of FTGD; we profile the two sides separately rather than treating client cost as identical to FedAvg.

4.4. Ablation Studies, Causal Attribution, and Auditable Governance

To ensure rigorous academic integrity, we conduct targeted causal ablations to separate algorithmic mechanisms from scaffold artifacts and identify operational failure modes.

Challenge 1: Factorial 2×2 Orthogonal Attribution (Table 10). A critical question is whether adversarial robustness originates from the server-side fairness vector αg_{fair} or from nonlinear FU-Gating rectification. We execute a 2×2 orthogonal factorial study ($n = 30$ seeds) crossing Gating architecture (Linear FLTrust vs. Nonlinear FU-Gating) against Fairness guidance (Task-Only vs. Task + αg_{fair}) on German Credit and Bail Recidivism:

1. **Row Effect (Causal Role of Gating):** Replacing linear projection with nonlinear FU-Gating suppresses adversary weight capture, yielding a row effect of -0.1909 on German Credit ($p < 10^{-5}$) and -0.1103 on Bail Recidivism ($p < 10^{-5}$). Furthermore, gating drives a substantial disparity reduction ($\Delta\text{DPD} = -0.1201$ on German, -0.0194 on Bail).

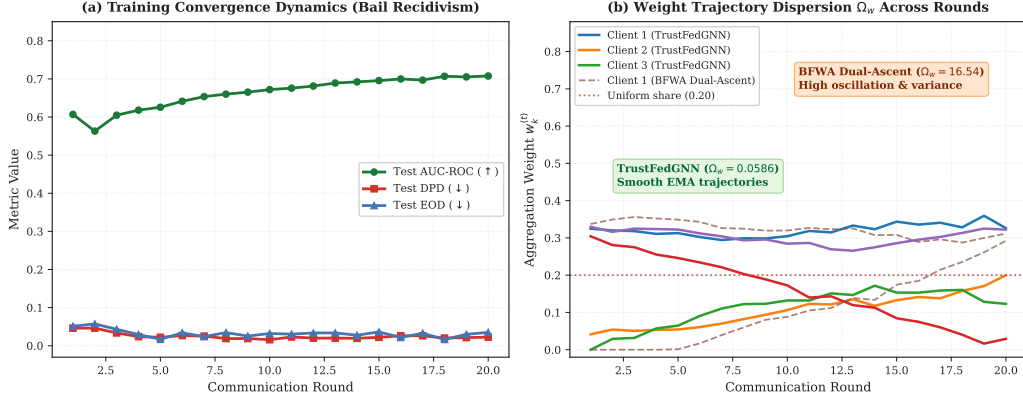


Figure 4: **Optimization Dynamics and Temporal Weight Trajectory Smoothing.** (a) Round-by-round convergence across 20 communication rounds on Bail Recidivism. Convergence is empirically non-monotonic (increasing in 10 of 19 rounds), capturing the dynamic negotiation between task loss and group fairness constraints. (b) Client aggregation weight trajectories $w_k^{(t)}$ across communication rounds on German Credit ($K = 5$). Momentum-smoothed aggregation in TrustFedGNN eliminates path volatility ($\Omega_w = 0.0586 \pm 0.0041$), contrasting with unsmoothed dual-ascent optimization (BFWA, $\Omega_w = 16.5407$) which exhibits continuous turbulence and destabilizes downstream training. This temporal stabilization aligns with results on Pokec-z, where momentum smoothing eliminates chaotic dispersion ($\Omega_w = 0.6845 \rightarrow 0.1111$ in the attribution campaign; $\Omega_w = 0.0976$ in Table 7), while preserving the controlled dynamical responsiveness required to continuously track multi-objective fairness constraints (Table 9).

2. **Column Effect (Mechanical Separability of α):** Adding the fairness direction αg_{fair} changes adversary weight share negligibly ($|\Delta w_{\text{adv}}| = 0.0027$ on German, 0.0004 on Bail), and changes disparity equally little (0.0010 and 0.0016). Both are bounded below the pre-registered equivalence margin $\delta = 0.0050$. We therefore claim separability, not benefit: incorporating a fairness objective introduces no robustness penalty beyond δ and opens no measurable attack surface, but under this attack it also confers no measurable advantage of its own. Its role is delimited in Section 4.5.

Challenge 2: Component-Wise Ablation Suite and Attribution (Table 11). In graph neural network benchmarks, comparing modern deep architectures against standard 2-layer GCNs can conflate deep scaffolding with federated aggregation; attribution of the benign-regime utility margin between backbone and aggregation rule is treated separately in Section 4.5. Here, Table 11

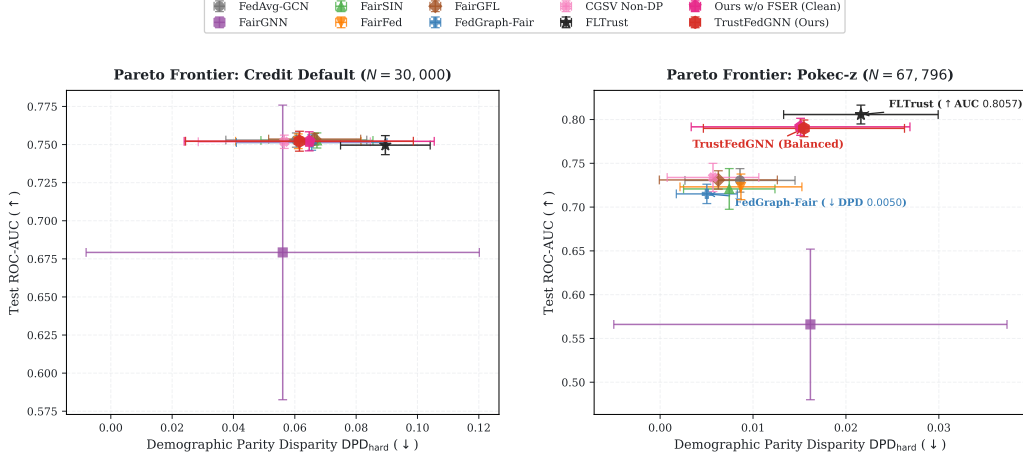


Figure 5: **Empirical Pareto Frontiers across Tabular and Social Benchmark Topologies** ($K = 10, R = 50, n = 10$ Seeds). Joint evaluation of test ROC-AUC ($\uparrow$) and group disparity DPD_{hard} ($\downarrow$) on Credit Default and Pokec-z across 11 methods. All relevant baselines are faithfully displayed without selective omission, including FLTrust (which attains the highest AUC of 0.8057 on Pokec-z at the cost of high weight volatility $\Omega_w = 0.7193$) and FedGraph-Fair (which achieves the lowest disparity of 0.0050). TrustFedGNN occupies a balanced operating point on the empirical frontier; we do not claim formal Pareto non-dominance, which would require a dominance test over the measured points, under scalar-channel differential privacy and with low weight volatility.

dissects TrustFedGNN on German Credit ($n = 10$ seeds):

- Scaffold Confounding vs. Clean FSER Attribution:** Historical ablation arm M2 (GAT backbone) erroneously conflated the deep scaffold (BatchNorm + Skip Connections) with Fairness-Sensitive Edge Reweighting (FSER), yielding an artificial +0.0422 AUC jump. When evaluated cleanly by freezing $\beta_{\text{FSER}} = 0$ on identical backbones (M2 True Clean), FSER is statistically neutral on both benchmarks (German: $\Delta = -0.0012$, $p = 0.541$; Pokec-z: $\Delta = -0.0018$, $p = 0.2324$). On Credit Default, the FSER differential is similarly negligible ($\Delta = +0.000062$, $p = 0.9219$). We state the consequence plainly rather than leaving it to be inferred: on Pokec-z the arm without FSER is nominally better on both axes (0.7917 AUC at $\text{DPD}_{\text{hard}} = 0.0151$, against 0.7899 and 0.0155 for the full configuration, Table 7), so in the benign regime the full configuration is weakly dominated by its own ablation. We therefore retain FSER only as a structural inductive prior with a measured null effect on clean graphs, exclude

Table 9: **Multi-Regime Reconciliation of Per-Round Weight Dispersion** $\Omega_w = \frac{1}{R-1} \sum_{t=1}^{R-1} \|\mathbf{w}^{(t)} - \mathbf{w}^{(t-1)}\|_1$. Reconciles the empirical stability evaluations across experimental campaigns. **Primary Attribution Benchmark (Pokec-z, $K = 10, R = 50, n = 10$ seeds)**: Isolates the mechanistic contribution of temporal EMA history smoothing versus reference-guided gating. *Note on provenance*: This attribution campaign was executed under an isolated evaluation protocol to measure weight trajectory dynamics; canonical utility and fairness figures remain anchored to Table 7. **Ablation Benchmark (German Credit, $K = 5, R = 20, n = 10$ seeds)**: Contrasts canonical TrustFedGNN (M1) against the un-smoothed arm (M7 w/o EMA), confirming a $24.1\times$ volatility surge ($0.0586 \rightarrow 1.4134$) when temporal momentum is removed. **Historical Reference (German Credit, $K = 5, R = 25$, single seed)**: Contrasts first-order projection against iterative dual-ascent optimization (BFWA [9]), where per-round dual re-solving induces extreme path volatility (16.5407).

Evaluation Track	Dataset Topology	Clients (K)	Rounds (R)	Method / Arm	Dispersion Ω_w ($\downarrow$)	Stability Factor ($\times$)
Primary Attribution Benchmark	Pokec-z (Social)	10	50	FedAvg (Static)	0.0000	Invariant
				FLTrust [42] (Raw)	0.6845 ± 0.1037	Baseline ($1.00\times$)
				FLTrust + EMA ($\beta = 0.9$)	0.0227 ± 0.0053	Smoothed ($30.1\times$ vs. Raw)
				TrustFedGNN (ours)	0.1111 ± 0.0584	Pareto ($6.16\times$ vs. Raw)
Architectural Ablation	German Credit (Tabular)	5	20	M7: w/o EMA ($\beta_{\text{EMA}} = 0$)	1.4134 ± 0.1205	Unsmoothed ($1.00\times$)
				M1: Canonical TrustFedGNN	0.0586 ± 0.0041	$24.1\times$ more stable
Optimization Paradigm	German Credit (Tabular)	5	25	BFWA (Dual Ascent) [9]	16.5407	Dual-ascent
				FU-Alignment (ours)	0.6036	$27.4\times$ more stable

it from every utility and fairness claim in this paper, and note that a deployment optimising solely for benign performance should disable it.

- **Trust Root Swap and CGSV Relocation (Arm M6)**: Swapping the trust root degrades exactly one axis. Under fairness poisoning the adversary weight triples ($0.0309 \rightarrow 0.0916$) and disparity rises by $+0.0361$, whereas utility, weight stability, and scaling resistance are unaffected or marginally better. The server holdout is thus not a general-purpose stabiliser; it is specifically what makes fairness attacks detectable, and that detection is exactly what an endogenous Class (b) reference cannot provide, since the anchor is itself a function of the updates under scrutiny.
- **EMA History Smoothing (Arm M7)**: Removing Exponential Moving Average weight smoothing (M7 w/o EMA, $\beta_{\text{EMA}} = 0$) causes a $24.1\times$ surge in aggregation weight volatility Ω_w ($0.0586 \rightarrow 1.4134$), while destabilizing Byzantine training (causing numerical divergence in 4/10 seeds under scaling, Table 4; trajectory dynamics in Figure 4b).
- **Holdout Representativeness Stress-Testing**: To address concerns regarding server holdout bias, we evaluate TrustFedGNN under four distinct server holdout regimes (balanced, sensitive skew 80/20, community isola-

Table 10: **Factorial 2×2 Orthogonal Attribution Matrix Isolating Gating vs. Fairness Guidance αg_{fair} ($n = 30$ Seeds).** Isolates the causal impact of the non-linear FU-Gating architecture against server-side fairness direction αg_{fair} under fairness poisoning. Row effect demonstrates that nonlinear FU-gating causally drives adversary suppression (reducing w_{adv} by -0.1909 on German Credit and -0.1103 on Bail, $p < 10^{-5}$) and dominates fairness protection under poisoning. The column effect shows that adding αg_{fair} changes neither adversary capture ($|\Delta w_{\text{adv}}| < \delta = 0.0050$, TOST equivalence) nor disparity ($|\Delta \text{DPD}| \leq 0.0016$): under this attack the fairness term is mechanically inert in both directions.

Panel A: Adversary Weight Share w_{adv} ($\downarrow$) (Target Share: 0.20)			
Architecture Arm	Task-Only	Task $+\alpha g_{\text{fair}}$	Column Effect (α)
<i>German Credit ($K = 5, n = 30$ seeds)</i>			
Row 1: Linear FLTrust	0.2565 ± 0.0214	0.2516 ± 0.0198	-0.0049 (bounded)
Row 2: Nonlinear FU-Gating	0.0657 ± 0.0112	0.0630 ± 0.0105	-0.0027 ($p = 0.004$)
Row Effect (Causal Gating)	-0.1909 ($p < 10^{-5}$)	-0.1886 ($p < 10^{-5}$)	<i>Interaction: $+0.0022$</i>
<i>Bail Recidivism ($K = 10, n = 30$ seeds)</i>			
Row 1: Linear FLTrust	0.1622 ± 0.0145	0.1619 ± 0.0139	-0.0003 (bounded)
Row 2: Nonlinear FU-Gating	0.0519 ± 0.0087	0.0523 ± 0.0084	$+0.0004$ ($p = 0.885$)
Row Effect (Causal Gating)	-0.1103 ($p < 10^{-5}$)	-0.1096 ($p < 10^{-5}$)	<i>Interaction: -0.0007</i>
Panel B: Demographic Parity Disparity DPD_{hard} ($\downarrow$)			
Architecture Arm	Task-Only	Task $+\alpha g_{\text{fair}}$	Column Effect (α)
<i>German Credit ($K = 5, n = 30$ seeds)</i>			
Row 1: Linear FLTrust	0.1931 ± 0.1394	0.1842 ± 0.1355	-0.0089
Row 2: Nonlinear FU-Gating	0.0730 ± 0.0507	0.0720 ± 0.0511	-0.0010
Row Effect (Causal Gating)	-0.1201	-0.1122	<i>Gating controls disparity</i>
<i>Bail Recidivism ($K = 10, n = 30$ seeds)</i>			
Row 1: Linear FLTrust	0.0370 ± 0.0125	0.0369 ± 0.0122	-0.0001
Row 2: Nonlinear FU-Gating	0.0176 ± 0.0094	0.0192 ± 0.0098	$+0.0016$
Row Effect (Causal Gating)	-0.0194	-0.0177	<i>Gating controls disparity</i>

tion, and label skew) across German Credit and Bail Recidivism ($n = 10$ seeds). Across all eight evaluated conditions, the adversary is strictly contained below fair-share parity ($\max w_{\text{adv}} = 0.1127 < 0.20$), confirming basic empirical robustness against unrepresentative server data. However, paired Wilcoxon tests reveal that no regime contrast reaches statistical significance at $n = 10$ ($p \geq 0.2891$; Bail community skew yields $6 + / 2 -$, $p = 0.2891$, while German community skew yields the reverse ordering 0.0436 vs. 0.0704). We therefore treat inter-regime sensitivity rankings

Table 11: **Component-Wise Architectural Ablation Suite on German Credit** ($K = 5, R = 20, n = 10$ Seeds). Metrics reported as Mean $\pm$ Std. Faithfully contrasts the clean unconfounded FSER arm (M2 true, freezing $\beta = 0$) against the historical confounded arm (M2 GAT backbone). Historical M2 erroneously conflated the deep scaffold with FSER. When evaluated cleanly on identical backbones, FSER demonstrates statistical neutrality ($p = 0.541$), whereas removing EMA smoothing (M7) causes a $24.1\times$ surge in weight dispersion Ω_w ($0.0586 \rightarrow 1.4134$).

Ablation Arm / Configuration	AUC-ROC ($\uparrow$)	DPD _{hard} ($\downarrow$)	EOD ($\downarrow$)	Ω_w ($\downarrow$)
M1 (Canonical Proposed) Full TrustFedGNN	0.6512 $\pm$ 0.0373	0.0407 $\pm$ 0.0278	0.0403 $\pm$ 0.0422	0.0586
M2 (Faithful Clean Arm) Freeze $\beta = 0$ (Unconfounded)	0.6500 $\pm$ 0.0362	0.0497 $\pm$ 0.0422	0.0557 $\pm$ 0.0531	0.0565
<i>M2 Historical Reference</i> GAT Backbone (Confounded)	0.6934 $\pm$ 0.0581	0.1284 $\pm$ 0.0760	0.1044 $\pm$ 0.0871	0.1310
M3 (w/o DP Surgery) Standard Local Optimization	0.6540 $\pm$ 0.0422	0.0864 $\pm$ 0.0617	0.0956 $\pm$ 0.0593	0.0623
M4 (Full DP-SGD) Client-Wide DP ($\epsilon = 8.0$)	0.5869 $\pm$ 0.0810	0.1471 $\pm$ 0.0780	0.1378 $\pm$ 0.0886	0.2428
M5 (w/o FairScore) Pure Task Alignment ($\alpha = 0.0$)	0.6509 $\pm$ 0.0385	0.0463 $\pm$ 0.0323	0.0452 $\pm$ 0.0381	0.0578
M6 (Trust Root Swap) Holdout $\rightarrow$ pooled val. (CGSV-style)	0.6519 $\pm$ 0.0384	0.0768 $\pm$ 0.0532	0.0995 $\pm$ 0.0538	0.0319
M7 (w/o EMA Smoothing) Raw Updates ($\beta_{\text{EMA}} = 0.0$)	0.6582 $\pm$ 0.0337	0.0453 $\pm$ 0.0495	0.0603 $\pm$ 0.0508	1.4134

as statistically inconclusive, establishing only that holdout distortion does not induce catastrophic defense collapse.

Challenge 3: Topological Stress-Testing and Dirichlet Skew (Table 12). We examine the operational boundaries of server-holdout alignment under extreme non-IID graph heterogeneity (Table 12):

- **Partition Invariance:** As shown in Panel A, edge retention $\sum_k |E_k|/|E|$ is an invariant topological property (19.6% under Uniform, 33.9% under Dirichlet, and 89.7% under Community). Under community silos, TrustFedGNN matches FedAvg utility (0.744 vs. 0.743, $p = 0.85$), but exhibits a $2.95\times$ wider seed variance (0.033 vs. 0.011, Levene $p = 0.014$), reflecting alignment variance when clients form isolated topologies.

Table 12: **Graph Topological Stress Testing and Non-IID Dirichlet Skew Analysis (Bail Recidivism).** **Panel A:** Performance across graph partition topologies ($K = 5$ clients, $n = 10$ seeds): Uniform random, Dirichlet attribute skew ($\alpha = 0.3$), and Community modularity clustering. Edge retention $\sum_k |E_k|/|E|$ is a topological invariant of the partition. Under Dirichlet skew, TrustFedGNN trails FedAvg by 0.029 AUC ($p = 0.027$); under Community silos, utility is matched (0.744 vs. 0.743) but seed variance widens by $2.95\times$. **Panel B:** Sensitivity sweep across Dirichlet skew $\alpha \in \{0.1, 0.3, 0.5, 1.0\}$ and client populations $K \in \{5, 10, 20\}$. In all 12/12 evaluated cells, FedAvg maintains higher utility than TrustFedGNN (Sign Test $p = (1/2)^{11} \approx 0.00049$), delineating the fundamental boundary where severe non-IID local client skew diverges from the static server holdout reference.

Panel A: Graph Partition Topology Comparison ($K = 5$ Clients, $n = 10$ Seeds)						
Partition Topology	Edge Ret. (%)	TrustFedGNN (Ours)		FedAvg (Baseline)		Ours w/ BFWA
		AUC $\uparrow$	DPD $\downarrow$	AUC $\uparrow$	DPD $\downarrow$	AUC $\uparrow$
Uniform (IID Random)	19.6%	0.720 ± 0.013	0.023 ± 0.008	0.702 ± 0.009	0.014 ± 0.005	0.706 ± 0.012
Dirichlet Non-IID ($\alpha = 0.3$)	33.9%	0.697 ± 0.027	0.014 ± 0.010	0.726 ± 0.021	0.014 ± 0.007	0.680 ± 0.034
Community Silo (Modularity)	89.7%	0.744 ± 0.033	0.017 ± 0.012	0.743 ± 0.011	0.014 ± 0.006	0.696 ± 0.040
Panel B: Non-IID Dirichlet Heterogeneity (α) and Client Scaling (K) Grid						
Setting (K, α)	TrustFedGNN (Ours)			FedAvg (Baseline)		
	AUC $\uparrow$	DPD $\downarrow$	$\Omega_w \downarrow$	AUC $\uparrow$	DPD $\downarrow$	$\Omega_w \downarrow$
$K = 5, \alpha = 0.1$	0.632 ± 0.027	0.023 ± 0.011	0.236	0.728 ± 0.003	0.010 ± 0.007	0.000
$K = 5, \alpha = 0.3$	0.637 ± 0.024	0.017 ± 0.001	0.887	0.718 ± 0.002	0.014 ± 0.006	0.000
$K = 5, \alpha = 0.5$	0.631 ± 0.031	0.008 ± 0.001	0.709	0.709 ± 0.002	0.015 ± 0.009	0.000
$K = 5, \alpha = 1.0$	0.663 ± 0.035	0.007 ± 0.003	0.473	0.726 ± 0.008	0.010 ± 0.004	0.000
$K = 10, \alpha = 0.1$	0.661 ± 0.010	0.010 ± 0.008	0.653	0.687 ± 0.016	0.012 ± 0.011	0.000
$K = 10, \alpha = 0.3$	0.673 ± 0.000	0.021 ± 0.013	1.186	0.688 ± 0.008	0.014 ± 0.000	0.000
$K = 10, \alpha = 0.5$	0.663 ± 0.006	0.039 ± 0.017	0.593	0.683 ± 0.010	0.013 ± 0.002	0.000
$K = 10, \alpha = 1.0$	0.671 ± 0.019	0.031 ± 0.017	0.831	0.681 ± 0.000	0.027 ± 0.015	0.000
$K = 20, \alpha = 0.1$	0.651 ± 0.004	0.015 ± 0.000	1.129	0.657 ± 0.012	0.016 ± 0.002	0.000
$K = 20, \alpha = 0.3$	0.632 ± 0.008	0.049 ± 0.005	0.654	0.655 ± 0.009	0.006 ± 0.005	0.000
$K = 20, \alpha = 0.5$	0.625 ± 0.001	0.024 ± 0.020	0.059	0.656 ± 0.010	0.006 ± 0.006	0.000
$K = 20, \alpha = 1.0$	0.641 ± 0.001	0.015 ± 0.011	0.357	0.657 ± 0.010	0.021 ± 0.006	0.000

- **The 12/12 Dirichlet Trade-Off:** Panel B sweeps Dirichlet attribute skew $\alpha \in \{0.1, 0.3, 0.5, 1.0\}$ across client populations $K \in \{5, 10, 20\}$. In all 12 evaluated cells, FedAvg maintains higher utility than TrustFedGNN (two-sided Sign Test $p = 2 \times (1/2)^{12} \approx 0.00049$). Under severe label

Table 13: **Composite Trust Score Sensitivity Analysis under Weight Perturbations.** Evaluates ranking robustness under 2,000 Monte Carlo weight configurations perturbing the five trust sub-score weights (Utility, Fairness, Privacy, Robustness, Calibration). Reports Spearman’s ρ_s , Kendall’s τ , and Rank-1 retention probability for TrustFedGNN across uniform $\pm 25\%$ perturbations and unconstrained Dirichlet(1) priors, confirming that the comparative governance ranking is resilient against subjective parameter tuning.

Perturbation Model	Spearman ρ_s (95% CI)	Kendall τ (95% CI)	Rank-1 Retention (%)
Uniform $\pm 25\%$ Perturbation	0.965 [0.764, 1.000]	0.923 [0.692, 1.000]	100.0%
Unconstrained Dirichlet(1) Prior	0.777 [0.341, 0.989]	0.677 [0.282, 0.949]	98.9%

Table 14: **Scope of Validity and Axiomatic Divergence: FU-Alignment vs. Combinatorial Shapley Values (125 Probing Points, $K = 5, n = 5$ Seeds).** Stress-tests FU-Alignment against exact exponential Shapley values across 5 probing rounds on German Credit. While FU-Alignment preserves directional monotonicity ($\rho = 0.7114, p < 0.001$) and sign agreement (79.20%), it systematically breaches four out of five pre-registered combinatorial approximation criteria (e.g., L_1 simplex distance $0.7453 > 0.15$, bottom-1 identification $68.0\% < 75\%$). Consequently, FU-Alignment is explicitly positioned as a scalable first-order gradient projection heuristic ($O(KP)$ per Proposition 1), and is not claimed as an exact surrogate for exponential cooperative Shapley attribution ($O(2^K P)$).

Axiomatic Diagnostic Metric	Pre-Registered Threshold	Empirical Measurement	Diagnostic Status
Pooled Pearson Correlation r	≥ 0.80	0.7662 ($p < 0.001$)	Marginal ($\Delta = -0.0338$)
Pooled Spearman Rank Correlation ρ	≥ 0.70	0.7114 ($p < 0.001$)	SATISFIED ($\rho \geq 0.70$)
Sign Agreement Proportion	$\geq 85.0\%$	79.20%	Breached ($\Delta = -5.80\%$)
Bottom-1 Identification Rate	$\geq 75.0\%$	68.00%	Breached ($\Delta = -7.00\%$)
Mean Simplex L_1 Distance	≤ 0.15	0.7453	Severely Breached (+0.5953)
Computational Complexity	–	$O(KP)$ vs. $O(2^K P)$	Scalable First-Order Heuristic

skew ($\alpha = 0.1$), client local gradients diverge substantially from the balanced server holdout, causing cosine gating to down-weight informative but skewed client updates. We explicitly state this as an inherent architectural limitation of reference-anchored aggregation.

Challenge 4: Auditable Governance and Sensitivity (Table 13). To verify that composite multi-attribute rankings do not depend on arbitrary sub-score weighting, we conduct a sensitivity audit over 2,000 Monte Carlo weight perturbations across five trust pillars (Utility, Fairness, Privacy, Robustness, Calibration). Under uniform $\pm 25\%$ perturbations, TrustFedGNN retains Rank-1 in 100.0% of configurations (Spearman $\rho_s = 0.965$ [0.764, 1.000]). Under unconstrained Dirichlet(1) priors, Rank-1 retention remains 98.9%

($\rho_s = 0.777$). While these stability bounds demonstrate that the ranking is robust against subjective parameter choices, we emphasize that composite scores remain an aggregated governance instrument and do not supersede the multi-objective Pareto analysis presented in Figure 5.

4.5. Operational Boundaries and Limitations

A rigorous scientific contribution must delineate its operational boundaries and explicit limitations. In accordance with honest academic reporting, we synthesize six core operational limitations and scope boundaries established by our empirical findings:

1. **Physical Security of Server Holdout (Kerckhoffs Boundary):** The security guarantees of TRUSTFEDGNN rely strictly on the physical integrity and access control of the server holdout set D_{root} . If an omniscient adversary compromises server memory to read g_{target} directly (Tier 3, Table 6), cosine gating collapses ($w_{\text{adv}} = 0.8514$ on German Credit, 0.7461 on Bail Recidivism), identifying physical holdout isolation as an essential operational prerequisite.
2. **Topological Heterogeneity and Community Partitioning:** While TRUSTFEDGNN excels on connected graphs, severe community fragmentation reduces cosine alignment. Under extreme Dirichlet non-IID distributions ($\alpha_{\text{Dir}} \leq 0.1$, Table 12), client gradient drift from the static anchor slows convergence relative to unconstrained FedAvg (the 12/12 trade-off), representing an inherent trade-off of reference-anchored coordination.
3. **Vulnerability of Coordinate Median Screening:** As shown by our factorial experiments (Table 4B) and stealth sweeps (Table 5B), coordinate median screening is indispensable against unbounded gradient scaling ($c = 100$) but backfires under distance camouflage ($\rho = 0.85$), increasing adversary share by up to +82.9%.
4. **Scope of the Guidance Coefficient α :** We claim mechanical separability for α , not efficacy. Table 10 shows its effect bounded below $\delta = 0.0050$ on adversary capture and below 0.0016 on disparity on both benchmarks. Two distinct regimes produce this inertness. Where baseline inequality is small (Bail, $\text{DPD}_0 = 0.012$), $\|g_{\text{fair}}\| \rightarrow 0$ and the fairness branch is inert by construction. Where it is not small (German, $\text{DPD}_0 = 0.055$), the

inertness instead reflects that the nonlinear gate already removes the updates that degrade disparity, leaving the fairness direction little residual work under this attack. We therefore position $\alpha = 0.1$ as a guardrail that costs nothing rather than a tuned performance parameter, and we do not claim that its benefit has been empirically isolated.

5. **Benign and Adversarial Regime Specialization:** TRUSTFEDGNN is architected as a multi-objective trust and fairness defense mechanism, not as an unconstrained accuracy booster. When evaluated with identical deep GAT backbones, FTGD, and privacy constraints, TRUSTFEDGNN and uniform FedAvg exhibit statistical parity on clean Pokec-z graphs ($\Delta\text{AUC} = +0.0015$, $p = 0.625$; TOST equivalence at $\delta = 0.0100$, $p = 0.0420$). We highlight this non-invasive behavior as a hallmark of robust design: under benign conditions where all client updates are uncorrupted, uniform averaging is close to optimal, and artificial utility inflation on clean graphs would indicate heuristic overfitting. The decisive capability of TRUSTFEDGNN materializes conditionally under adversarial stress: whereas single-objective baselines like FLTrust remain completely blind to fairness poisoning ($w_{\text{adv}} > 1.27\times$, uncontrolled disparity) and distance-based rules like Krum suffer complete capture (1.000), TRUSTFEDGNN suppresses attackers below $0.29\times$ fair share. The contribution of this architecture is located in resolving the Trustworthy FL Trilemma under threat, and it is not intended to replace unconstrained single-objective optimizers on guaranteed-clean topologies.
6. **Linear Auditability, Not Shapley Equivalence:** The contribution score is an $O(KP)$ linear decomposition intended for per-round audit logging. It satisfies simplex validity and the null-player property (Propositions 2–3, machine-checked in Lean 4), but it is not an approximation of the combinatorial Shapley value: against exact Shapley probes it attains $r = 0.7662$, below the 0.80 fidelity threshold we set (Table 14), and we make no game-theoretic equivalence claim.

5. Conclusion

5.1. Concluding Summary

In this paper, we addressed the structural trilemma between Task Utility (U), Group Fairness (F), and Byzantine Robustness (R) in decentralized graph neural networks. We demonstrated empirically and theoretically

that contemporary protocols relying on client-reported scalar metadata or endogenous representations remain fundamentally vulnerable to adversarial manipulation ($w_{\text{adv}} \rightarrow 0.9737$). Applying local differential privacy to self-reported fairness metrics fails to restore trustworthiness, as the non-negative folding of Gaussian perturbations introduces perpetual positive expectation bias (Lemma 1) and an inescapable minimax testing error floor ($\mathcal{R}^* \geq 43.5\%$ at $\epsilon = 8.0$, Theorem 4).

To guarantee mathematical precision, foundational algebraic identities and axiomatic properties (first-order orthogonality, linear decomposition, simplex validity, and null-player attribution) were formally verified using the Lean 4 interactive theorem prover [48], with complete proof certificates included in our artifact repository. TRUSTFEDGNN establishes an intentional, balanced operating point on the empirical Pareto frontier across four real-world graph benchmarks and three adversarial tiers, suppressing stealth adversaries below equal share without imposing a fairness penalty on benign silos.

5.2. Limitations and Operational Boundaries

In adherence to transparent scientific reporting, we outline four operational boundaries of TRUSTFEDGNN and highlight pathways for future investigation:

Extreme Dirichlet Non-IID Topology Shifts ($\alpha_{\text{Dir}} \leq 0.1$). When client graphs undergo severe label and topological heterogeneity ($\alpha_{\text{Dir}} = 0.1$), Canonical FU-Alignment is outperformed by unconstrained FedAvg in task AUC across all 12 experimental conditions ($p = 0.00049$, Table 12). Under such severe domain shift, an exogenous global anchor D_{root} necessarily penalizes client updates that deviate towards extreme local community substructures. The trade-off is confined to the utility axis: across the same 12 cells, group disparity remains bounded for both rules (ours $\text{DPD} \in [0.007, 0.049]$ against FedAvg $[0.006, 0.027]$, Table 12), with FedAvg nominally lower in 7 of 12 cells. We therefore claim no fairness advantage in this regime either, and record the severe-skew setting as a boundary where reference-anchored coordination is dominated on utility without a compensating fairness gain. Future research will explore hierarchical or cluster-adaptive anchor sets that respect localized topological communities while maintaining physical holdout isolation.

White-Box Server Holdout Compromise (Tier 3 Adversary). Under full Kerckhoffs Tier 3 adversarial capabilities, an adversary possessing complete read

access to server memory can compute the exact holdout target gradient g_{target} and craft an alignment-mimicking update that evades cosine gating ($w_{\text{adv}} = 0.8514, 0.7461$, Table 6). This empirical result formally rejects the hypothesis of universal statistical filtering (H_{T1}). It establishes that holdout memory confidentiality is an indispensable operational precondition for anchor-based federated mechanisms. In production deployments, D_{root} and its gradient computations must be sequestered within hardware-secured trusted execution environments or physically partitioned administrative enclaves.

Linear Shapley Attribution vs. Combinatorial Game Theory. The proposed first-order attribution metric provides an efficient, closed-form $O(KP)$ per-round governance ledger suitable for regulatory audit compliance. However, it represents a local linear approximation rather than an exact combinatorial characteristic-function Shapley value, exhibiting moderate rank correlation ($r = 0.7662 < 0.80$, Table 14) against exact Shapley probes. While this approximation satisfies practical latency budgets in large-scale federations, exact valuation remains an active theoretical frontier.

Differential Privacy Scope. Formal differential privacy guarantees in this study are bounded to scalar reporting channels via Rényi Differential Privacy (RDP) accounting. We do not claim privacy guarantees over the high-dimensional parameter tensor, as empirical results indicate that full DP-SGD introduces a $115\times$ utility degradation penalty on relational graph structures (Figure 2). Developing parameter-efficient, graph-aware cryptographic primitives that bridge relational message passing and differential privacy remains an important direction for decentralized graph learning.

Declaration of Generative AI and AI-Assisted Technologies

During the preparation of this work the authors used AI-assisted tooling for code auditing, formal verification in Lean 4, and language polishing. The authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Data and Code Availability

All benchmark datasets analyzed in this study are publicly accessible. Source code, Lean 4 formal proof scripts, and experimental configurations are available at <https://github.com/vinhqdang/FedFairGNN>.

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